Package ‘sautilities’

July 1, 2026

Title Seasonal Adjustment Utilities For Use With the Seasonal Package

Version 6.0

Description Several utilities to provide support for the seasonal package. This includees routines that select the X-11 seasonal filter based on the magnitude of the estimate of the seasonal moving average coefficient from the airline model, duplicates the functionality of the TERROR software that performs quality control on time series based on one step ahead forecasts, generate model summaries from seas objects, generate names and abbreviations for X-13ARIMA-SEATS tables, save spec files, seasonal objects, and metafiles into external files, process list objects of numbers, indicate which elements of a list have try-errors, replace NA with a string, set outlier critical values, add an outlier spec to a static seas element, get indexes and entries from UDG output generated from seasonal, save seasonal objects into R scripts and X-13ARIMA-SEATS spec files, and functions to collect diagnostics summaries for various X-13ARIMA-SEATS diagnostics, and read in files output by the regCMPNT program.

License MIT + file LICENSE

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

RoxygenNote 8.0.0

Depends R (>= 3.6)

Imports lubridate,
Rdpack,
seasonal (>= 1.10.0),
stringr,
stringi

RdMacros Rdpack

Contents

`absmax` *Maximum absolute value of a vector*

Description

Generates the maximum of the absolute value of a numeric vector.

Usage

```
absmax(x = NULL, return_abs = TRUE)
```

Arguments

`x`	Vector of numbers. This is a required element.
`return_abs`	Logical scalar; if `TRUE`, return the absolute maximum value. If `FALSE`, maintain the sign of the original value. Default is `TRUE`.

Details

Version 2.1, 2026-06-04

Value

Maximum of the absolute value of a vector.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
r50 <- rnorm(50)
r50.absmax <- absmax(r50)
```

`absolute_pct_diff` *Generate the either the average or median absolute percentage difference between two vectors*

Description

Generate the average absolute percentage difference (AAPD) or median absolute percentage difference (MAPD) between two vectors - the vectors must be the same length.

Usage

```
absolute_pct_diff(x1 = NULL, x2 = NULL, use_median = FALSE)
```

Arguments

x1	numeric vector; This is a required entry.
x2	numeric vector; This is a required entry, and must be the same length as x1.
use_median	logical scalar, if TRUE, returns the median of the absolute percentage difference; if FALSE, returns the average of the absolute percentage difference Default is FALSE.

Details

Version 2.0, 6/21/2024

Value

Either the average or median absolute percentage difference of the two series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
                  x11="", transform.function = "log", forecast.maxlead=20,
                  check.print = c( "pacf", "pacfplot" ))
ukgas_seats_seas <-
   seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
                  transform.function = "log",, forecast.maxlead=20,
                  check.print = c( "pacf", "pacfplot" ))
ukgas_x11_sa     <- seasonal::final(ukgas_seas)
ukgas_seats_sa   <- seasonal::final(ukgas_seats_seas)
ukgas_sa_aapd    <-
   sautilities::absolute_pct_diff(ukgas_seats_sa, ukgas_x11_sa)
ukgas_sa_mapd    <-
   sautilities::absolute_pct_diff(ukgas_seats_sa, ukgas_x11_sa, use_median = TRUE)
```

`acf_fail` *ACF Test failure message*

Description

Tests whether the sample autocorrelation of the residuals from a time series model fails the Ljung-Box (Ljung and Box 1978) or Box-Pierce Q test. (Box and Pierce 1970).

Usage

```
acf_fail(
  udg_list = NULL,
  acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  num_sig = 8,
  include_pacf = TRUE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package. (Sax and Eddelbuettel 2018). This is a required entry.
`acf_lags_fail`	Lags of the ACF to test. Default is `c(1, 2, 3, 4, 8)`.
`num_sig`	Limit for number of lags with significant ACF values. Default is `4`.
`include_pacf`	Logical scalar that indicates if the PACF is included in the testing. Default is `TRUE`

Details

Version 4.0, 2026-2-24

Value

Logical object which is `TRUE` if series fails the ACF test, `FALSE` otherwise.

Author(s)

Brian C. Monsell, <`monsell.brian@bls.gov`> or <`bcmonsell@gmail.com`>

References

Box GEP, Pierce DA (1970). “Distribution of Residual Autocorrelations in Autoregressive Integrated Moving Average Time Series Models.” *Journal of the American Statistical Association*, **65**, 1509–1526. doi:10.1080/01621459.1970.10481180.

Ljung GM, Box GEP (1978). “On a Measure of Lack of Fit in Time Series Models.” *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_acf_fail <-
     acf_fail(ukgas_udg,
              acf_lags_fail = c(1, 2, 3, 4, 8),
              num_sig = 4)
```

---

acf_fail_why *ACF Test Explanation*

Description

Generates text on why the sample autocorrelation of the residuals from a time series model fails the Ljung-Box (Ljung and Box 1978) or Box-Pierce Q test. (Box and Pierce 1970).

Usage

```
acf_fail_why(
  udg_list = NULL,
  acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  num_sig = 8,
  include_pacf = TRUE,
  return_both = FALSE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`acf_lags_fail`	Integer vector; lags of the ACF to test. Default is `c(1, 2, 3, 4, 12, 24)`.
`num_sig`	Integer scalar; limit for number of lags with significant ACF values. Default is `8`.
`include_pacf`	Logical scalar that indicates if the PACF is included in the testing. Default is `TRUE`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or just produce a warning. Default is `FALSE`.

Details

Version 5.1, 2026-03-06

Value

Character object that tells why series fails the ACF test, `'pass'` otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Box GEP, Pierce DA (1970). "Distribution of Residual Autocorrelations in Autoregressive Integrated Moving Average Time Series Models." *Journal of the American Statistical Association*, **65**, 1509–1526. doi:10.1080/01621459.1970.10481180.

Ljung GM, Box GEP (1978). "On a Measure of Lack of Fit in Time Series Models." *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_acf_fail_why <-
     acf_fail_why(ukgas_udg, acf_lags_fail = c(1, 2, 3, 4, 8),
                  num_sig = 4, return_both = TRUE)
```

---

acf_test *Global ACF test*

Description

Tests whether the residuals from a time series model has acceptable autocorrelation in the residuals.

Usage

```
acf_test(
  seas_obj = NULL,
  num_sig = 8,
  acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  acf_lags_warn = c(12, 24),
  include_pacf = TRUE,
  return_this = "test"
)
```

Arguments

`seas_obj`	A seas object generated by `seas()` of the `seasonal` package. (Sax and Eddelbuettel 2018). This is a required entry.
`num_sig`	Limit for number of lags with significant ACF values. Default is `8`.
`acf_lags_fail`	Integer vector; lags of the ACF to test. Default is `c(1, 2, 3, 4, 12, 24)`.
`acf_lags_warn`	Integer vector; lags of the ACF to test for warnings. Default is `c(12, 24)`.
`include_pacf`	Logical scalar that indicates if the PACF is included in the testing. Default is `TRUE`.
`return_this`	character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 5.1, 2026-03-06

Value

A text string denoting if series passes, fails, or has a warning for residual autocorrelation. If model diagnostics not found, return `'none'`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_acf_test <-
   acf_test(ukgas_seas,
      num_sig = 4,
      acf_lags_fail = c(1, 2, 3, 4, 8),
      acf_lags_warn = c(4, 8),
      return_this = "both")
```

acf_warn *ACF test warning message*

Description

Tests whether the residuals from a time series model generates a warning for the ACF test.

Usage

```
acf_warn(udg_list = NULL, acf_lags_warn = c(12, 24))
```

Arguments

udg_list	List object generated by udg() function of the seasonal package. (Sax and Eddelbuettel 2018). This is a required entry.
acf_lags_warn	Numeric vector; Lags of the ACF to test for warnings.

Details

Version 3.0, 2026-02-24

Value

Logical object which is TRUE if series generates a warning for the ACF test, FALSE otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_acf_warn <- acf_warn(ukgas_udg, acf_lags_warn = c(4,8))
```

`acf_warn_why` *ACF Test Warning Message*

Description

Generates text on why the sample autocorrelation of the residuals from a time series model fails the Ljung-Box or Box-Pierce Q test.

Usage

```
acf_warn_why(udg_list = NULL, acf_lags_warn = c(12, 24), return_both = FALSE)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`acf_lags_warn`	Integer vector; lags of the ACF to test for warnings. Default is `c(12, 24)`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 4.1, 2026-03-06

Value

Character string which tells why the series generates a warning for the ACF test, `'pass'` otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_acf_warn_why <-
     acf_warn_why(ukgas_udg, acf_lags_warn = c(4, 8), return_both = TRUE)
```

all_model_diag *Model diagnostic summary*

Description

Generate a summary of model diagnostics for a single series.

Usage

```
all_model_diag(
  seas_obj = NULL,
  add_aicc = FALSE,
  add_norm = FALSE,
  add_auto_out = FALSE,
  add_seasonal = FALSE,
  add_spec = FALSE,
  add_lbq = TRUE,
  set_lbq_lags_fail = c(12, 24),
  set_lbq_p_limit = 0.01,
  set_acf_num_sig = 8,
  set_acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  set_acf_lags_warn = c(12, 24),
  add_pacf = TRUE,
  qs_test_span = FALSE,
  set_qs_p_limit_pass = 0.01,
  set_qs_p_limit_warn = 0.05,
  set_qs_robust_sa = TRUE,
  set_spec_peak_level = 6,
  set_spec_peak_warn = 3,
  return_list = FALSE
)
```

Arguments

- `seas_obj` A seas object generated from a call of `seas` on a single time series (Sax and Eddelbuettel 2018). This is a required entry.
- `add_aicc` logical scalar; add AICC value (Burnham and Anderson 2004) to the summary. (Akaike 1973). Default is `TRUE`.
- `add_norm` logical scalar; add normality statistics to the summary. Default is `FALSE`.
- `add_auto_out` logical scalar; add identified automatic outliers to the summary. Default is `FALSE`.
- `add_seasonal` logical scalar; add QS test for seasonality to the summary. Default is `FALSE`.
- `add_spec` logical scalar; add test for spectral peaks to the summary. Default is `FALSE`.
- `add_lbq` logical scalar; add test for Ljung-Box Q to the summary (Ljung and Box 1978). Default is `TRUE`; if set to `FALSE`, a more extensive test of ACF will be done.
- `set_lbq_lags_fail` Integer vector; lags of the Ljung-Box Q to test. Default is `c(12, 24)`.
- `set_lbq_p_limit` Numeric scalar; P-value limit for Ljung-Box Q test. Default is `0.01`.

set_acf_num_sig
: Limit for number of lags with significant ACF values. Default is 8.

set_acf_lags_fail
: Integer vector; lags of the ACF to test Default is c(1, 2, 3, 4, 12, 24).

set_acf_lags_warn
: Integer vector; lags of the ACF to test for warnings. Default is c(12, 24).

add_pacf
: logical scalar; include PACF in ACF results. Default is TRUE.

qs_test_span
: logical scalar; test span of data in QS seasonal test rather than full series. For details on the QS test, see Section 7.17 of the X-13ARIMA-SEATS Referene Manual (U. S. Census Bureau 2025), Maravall (2012), and Bell et al. (2022). Default is FALSE.

set_qs_p_limit_pass
: Numeric scalar; P-value limit for QS statistic for passing QS test. Default is 0.01.

set_qs_p_limit_warn
: Numeric scalar; P-value limit for QS statistic for warning message. Default is 0.05.

set_qs_robust_sa
: Logical scalar indicating if original series adjusted for extremes is included in testing. Default is TRUE.

set_spec_peak_level
: Integer scalar; a limit to determine if a frequency has a spectral peak. Default is 6.

set_spec_peak_warn
: Integer scalar; a limit to produce a warning that a frequency may have a spectral peak. Default is 3.

return_list
: logical scalar; return a list rather than a vector. Default is FALSE.

Details

Version 6.5, 2026-03-09

Value

vector or list of model diagnostics for a given series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Akaike H (1973). "Information Theory and an Extension of the Likelihood Principle." In Petrov BN, Czaki F (eds.), *Second International Symposium on Information Theory*, 267-287. Akademia Kiado, Budapest.

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Burnham KP, Anderson DR (2004). "Multimodal Inference: Understanding AIC and BIC in Model Selection." *Sociological Methods and Research*, **33**, 261–304.

Ljung GM, Box GEP (1978). "On a Measure of Lack of Fit in Time Series Models." *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, `https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf`.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers, x11="",
                  slidingspans = "",
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  regression.aictest = "td",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  check.save = "acf")
air_diag <-
  all_model_diag(air_seas,
     add_seasonal = TRUE,
     add_aicc = TRUE,
     add_norm = TRUE,
     add_auto_out = TRUE,
     add_pacf = FALSE,
     qs_test_span = TRUE,
     return_list = TRUE)
```

`all_model_diag_list` *Model diagnostic summary from a list*

Description

Generate a summary of model diagnostics from a list of `seas` objects series. Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
all_model_diag_list(
  seas_obj_list = NULL,
  add_aicc = FALSE,
  add_norm = FALSE,
  add_auto_out = FALSE,
  add_seasonal = FALSE,
  add_spec = FALSE,
```

```
  add_lbq = TRUE,
  set_lbq_lags_fail = c(12, 24),
  set_lbq_p_limit = 0.01,
  set_acf_num_sig = 8,
  set_acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  set_acf_lags_warn = c(12, 24),
  add_pacf = TRUE,
  qs_test_span = FALSE,
  set_qs_p_limit_pass = 0.01,
  set_qs_p_limit_warn = 0.05,
  set_qs_robust_sa = TRUE,
  set_spec_peak_level = 6,
  set_spec_peak_warn = 3,
  return_data_frame = FALSE
)
```

Arguments

seas_obj_list
: List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.

add_aicc
: logical scalar; add AICC value (Burnham and Anderson 2004) to the summary. (Akaike 1973). Default is TRUE.

add_norm
: logical scalar; add normality statistics to the summary. Default is FALSE.

add_auto_out
: logical scalar; add identified automatic outliers to the summary. Default is FALSE.

add_seasonal
: logical scalar; add QS test for seasonality to the summary. For details on the QS test, see Section 7.17 of the X-13ARIMA-SEATS Referene Manual (U. S. Census Bureau 2025), and Maravall (2012). Default is FALSE.

add_spec
: logical scalar; add test for spectral peaks to the summary. Default is FALSE.

add_lbq
: logical scalar; add test for Ljung-Box Q to the summary (Ljung and Box 1978). Default is TRUE; if set to FALSE, a more extensive test of ACF will be done.

set_lbq_lags_fail
: Integer vector; lags of the Ljung-Box Q to test Default is c(12, 24).

set_lbq_p_limit
: Numeric scalar; P-value limit for Ljung-Box Q test Default is 0.01.

set_acf_num_sig
: Limit for number of lags with significant ACF values Default is 8.

set_acf_lags_fail
: Integer vector; lags of the ACF to test Default is c(1, 2, 3, 4, 12, 24).

set_acf_lags_warn
: Integer vector; lags of the ACF to test for warnings Default is c(12, 24).

add_pacf
: logical scalar; include PACF in ACF results. Default is TRUE.

qs_test_span
: logical scalar; test span of data in QS seasonal test rather than the full series. Default is FALSE.

set_qs_p_limit_pass
: Numeric scalar; P-value limit for QS statistic for passing. Default is 0.01.

set_qs_p_limit_warn
: Numeric scalar; P-value limit for QS statistic for warning messages. Default is 0.05.

set_qs_robust_sa
: Logical scalar indicating if original series adjusted for extremes is included in testing. Default is TRUE.

set_spec_peak_level
: Integer scalar; limit to determine if a frequency has a spectral peak. Default is 6.

set_spec_peak_warn
: Integer scalar; limit to produce a warning that a frequency may have a spectral peak. Default is 3.

return_data_frame
: logical scalar; if TRUE, return a data frame of the diagnostics, otherwise return a matrix. Default is FALSE.

Details

Version 9.3, 2026-03-06

Value

Matrix or data frame of model diagnostics for a given set of series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Akaike H (1973). "Information Theory and an Extension of the Likelihood Principle." In Petrov BN, Czaki F (eds.), *Second International Symposium on Information Theory*, 267-287. Akademia Kiado, Budapest.

Burnham KP, Anderson DR (2004). "Multimodal Inference: Understanding AIC and BIC in Model Selection." *Sociological Methods and Research*, **33**, 261–304.

Ljung GM, Box GEP (1978). "On a Measure of Lack of Fit in Time Series Models." *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
```

```
                   forecast.maxlead=36, x11 = "",
                   check.print = c( "pacf", "pacfplot" ),
                   check.save = "acf")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_diag <-
    all_model_diag_list(unemp_seas_update,
                        add_aicc = TRUE,
                        add_norm = TRUE,
                        add_auto_out = TRUE,
                        add_spec = TRUE,
                        add_pacf = FALSE,
                        qs_test_span = TRUE)
```

`check_stats` *Displays various X-13 diagnostics*

Description

Displays various X-13ARIMA-SEATS diagnostics for a single series (U. S. Census Bureau 2025). Seasonality diagnostics are described in Bell et al. (2022). Adapted from a script developed by Osbert Pang (US Census Bureau, osbert.c.pang@census.gov)

Usage

```
check_stats(
  seas_obj = NULL,
  print_summary = TRUE,
  test_full = TRUE,
  test_span = TRUE,
  acf_num_sig = 8,
  acf_lags_fail = c(1, 2, 3, 4, 12, 24),
  acf_lags_warn = c(12, 24),
  model_t_value = 3,
  model_p_value = 0.05,
  otl_auto_limit = 5,
  otl_all_limit = 5,
  d11f_p_level = 0.01,
  qs_p_limit_pass = 0.01,
  qs_p_limit_warn = 0.05,
  qs_p_limit_fail = 0.01,
  qs_robust_sa = TRUE,
  sf_limit = 25,
  change_limit = 40,
  mq_fail_limit = 1.2,
  mq_warn_limit = 0.8,
  return_list = FALSE
)
```

Arguments

`seas_obj` A `seas` object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.

print_summary
: Logical object; if TRUE, print the result of summary(seas_obj); if FALSE, a model summary will be printed out. Default is TRUE.

test_full
: Logical scalar indicating whether to apply the QS test to the full series span. Default is TRUE.

test_span
: Logical scalar indicating whether to test the QS test to the final 8-year span used by the spectrum diagnostic. Default is TRUE.

acf_num_sig
: Numeric object; limit for number of lags with significant ACF values. Default is 8.

acf_lags_fail
: Numeric vector; lags of the ACF to test. Default is c(1, 2, 3, 4, 12, 24).

acf_lags_warn
: Numeric vector; lags of the ACF to test for warnings, Default is c(12, 24).

model_t_value
: t-statistic limit for regressors. Default is 3.0.

model_p_value
: p-value limit for regressors. Default is 0.05.

otl_auto_limit
: limit for number of automatically identified outliers. Default is 5.

otl_all_limit
: limit for number of outlier regressors. Default is 5.

d11f_p_level
: p-level used to test the d11 f-test for residual seasonality. Default is 0.01.

qs_p_limit_pass
: Numeric scalar; P-value limit for QS statistic for passing. Default is 0.01.

qs_p_limit_warn
: Numeric scalar; P-value limit for QS statistic for warning. Default is 0.05.

qs_p_limit_fail
: Numeric scalar; P-value limit for QS statistic for failing. Default is 0.01.

qs_robust_sa
: Logical scalar indicating if original series adjusted for extremes is included in testing. Default is TRUE.

sf_limit
: Numeric object; limit for the percentage of seasonal spans flagged. Default is 25.

change_limit
: Numeric object; limit for the percentage of month-to-month changes flagged. Default is 40.

mq_fail_limit
: Numeric scalar; value above which the M or Q statistic fails. Default is 1.2.

mq_warn_limit
: Numeric scalar; value above which the M or Q statistic gives a warning message if it is less than this_fail_limit, Default is 0.8.

return_list
: Logical scalar; indicates if the function will return a summary of diagnostics. Default is TRUE.

Details

Version 4.2, 2026-03-05

Value

Displays assorted seasonal adjustment and modeling diagnostics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
     seasonal::seas(UKgas,
          series.period = 4,
          arima.model = "(0 1 1)(0 1 1)",
          x11="",
          transform.function = "log",
          forecast.maxlead=20,
          check.print = c( "pacf", "pacfplot" ))
ukgas_check_stats <-
   check_stats(ukgas_seas,
               acf_num_sig = 5,
               acf_lags_fail = c(1, 2, 3, 4, 8),
               acf_lags_warn = c(4, 8),
               otl_auto_limit = 4,
               otl_all_limit = 6,
               return_list = TRUE)
```

`check_table` *Check table name*

Description

Check X-13ARIMA-SEATS table name so that the B1 table can be extracted when no prior adjustment is done with the `seasonal` package (Sax and Eddelbuettel 2018).

Usage

```
check_table(seas_obj = NULL, this_series = NULL, to_upper = FALSE)
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_series`	Character string; a table name or abbreviation from X-13ARIMA-SEATS.
`to_upper`	Logical scalar; return table name that is uppercase. Default is `FALSE`.

Details

Version 1.3, 2026-04-16

Value

Character string; an X-13ARIMA-SEATS table name. If the table is "b1" or "adjoriginal", this function will return "a1".

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_short <-
   seasonal::seas(AirPassengers,
                  series.span = ',1960.9',
                  transform.function = "log",
                  arima.model="(0 1 1)(0 1 1)",
                  forecast.maxlead = 36,
                  x11 = "",
                  series.save = "b1",
                  slidingspans = "")
this_series <-
    check_table(air_seas_short, "b1")
```

---

choose_optimal_seasonal_filter
Choose Optimal X-11 seasonal moving average

Description

Choose the optimal X-11 seasonal moving average based on the value of the seasonal moving average coefficient from an airline model.

Usage

```
choose_optimal_seasonal_filter(
  this_seasonal_theta = NULL,
  dp_limits = TRUE,
  use_3x15 = TRUE
)
```

Arguments

`this_seasonal_theta`
: Numeric scalar; seasonal moving average coefficient from an airline model. This is a required entry.

`dp_limits`
: Logical scalar; if `TRUE` limits from Planas and Depoutot (2002) will be used to choose the moving average, else limits from Chu et al. (2012) will be used. Default is `TRUE`.

use_3x15 Logical scalar; if TRUE 3x15 seasonal filter will be returned if chosen, otherwise function will return a 3x9 value. Default is FALSE.

Details

Version 2.0, 2026-02-24

Value

The optimal X-11 seasonal filter.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Chu Y, Tiao GC, Bell WR (2012). "A Mean Squared Error Criterion for Comparing X-12-ARIMA and Model-Based Seasonal Adjustment Filters." *Taiwan Economic Forecast and Policy*, **43**(1), 1–32. https://www.census.gov/content/dam/Census/library/working-papers/2012/adrm/chutiaobelltefp.pdf.

Planas C, Depoutot R (2002). "Controlling Revisions in ARIMA-Model-Based Seasonal Adjustment." *Journal of Time Series Analysis*, **23**, 193–214. doi:10.1111/14679892.00262.

Examples

```
shoes_seas <-
   seasonal::seas(shoes2008,
                  x11="", slidingspans = "",
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  regression.aictest = c("td", "easter"),
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ))
shoes_seasonal_MA <- shoes_seas$est$coefficients[["MA-Seasonal-12"]]
this_seasonal  <-
    choose_optimal_seasonal_filter(shoes_seasonal_MA)
this_seasonal2 <-
    choose_optimal_seasonal_filter(shoes_seasonal_MA, dp_limits = FALSE)
```

---

combined_spectrum_test *Combined spectrum test from Maravall (2012)*

Description

Generate a test for seasonality by combining the results from the AR(30) and Tukey nonparametric spectrums as laid out in Maravall (2012).

Usage

```
combined_spectrum_test(
  seas_obj = NULL,
  this_ar_spec_cv = NULL,
  n_sim = 1e+05,
  this_series = "series.adjoriginal",
  take_log = TRUE,
  take_diff = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_ar_spec_cv	List object with two elements - 99 and 95 percent critical values for the frequencies of the AR(30) spectrum as generated by the gen_ar_spec_cv function.
n_sim	integer scalar; number of simulations; this option is used only if this_ar_spec_cv is not specified. Default is 100000.
this_series	character string; the table used to generate the AR(30) spectrum. Default is "series.adjoriginal".
take_log	logical scalar; indicates if the AR spectrum is generated from the log of the data. Default is TRUE.
take_diff	logical scalar; indicates if the data is differenced before the AR spectrum is generated. Default is TRUE.

Details

Version 4.3, 2026-04-21

Value

TRUE if spectral evidence of seasonality is detected; FALSE if not.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
    seasonal::seas(AirPassengers,
                   series.save = "b1",
                   transform.function = "log",
```

```
                    arima.model="(0 1 1)(0 1 1)",
                    forecast.maxlead = 36,
                    slidingspans = "")
this_spectrum_test <-
    combined_spectrum_test(air_seas, n_sim = 1000)
```

---

compare_dates *Date Match*

Description

Compare two dates to see if they match.

Usage

```
compare_dates(this_date = NULL, comp_date = NULL)
```

Arguments

this_date	Integer array of length 2, a date where the first element is the year and the second element is the month or quarter. This is a required entry.
comp_date	Integer array of length 2, a date to comapare to this_date. This is a required entry.

Details

Version 3.3, 2026-02-24

Value

Logical scalar; TRUE if the dates match, FALSE if they don't.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
match_start <- compare_dates(start(shoes2007), c(1990,1))
```

convert_date_string_to_date
Convert date string from UDG output

Description

Convert a date string from the output of the udg function from the seasonal package (Sax and Eddelbuettel 2018) to a c(year, month) date.

Usage

```
convert_date_string_to_date(this_date_string)
```

Arguments

this_date_string
Date string usually extracted from the X-13 UDG output.

Details

Version 2.0, 2026-02-24

Value

Integer array of length 2 with the year and month/quarter of from the date string.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
    seasonal::seas(AirPassengers,
                   series.save = "b1",
                   transform.function = "log",
                   arima.model="(0 1 1)(0 1 1)",
                   forecast.maxlead = 36,
                   slidingspans = "")
this_start_spec_string <- seasonal::udg(air_seas, "startspec")
this_start_spec        <-
    convert_date_string_to_date(this_start_spec_string)
```

`convert_dv_to_csv` *convert a* `datevalue` *formatted file into a csv file ready to be read by Python*

Description

Process a data file stored in `datevalue` form and produce a `.csv` file that is ready to be read by Python. For more information on the `datevalue` format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
convert_dv_to_csv(this_dv_file = NULL, this_csv_file = NULL, start_date = NULL)
```

Arguments

`this_dv_file`	Character string; name of file with a time series stored in `datevalue` format. This is a required argument.
`this_csv_file`	Character string; name of file time series array to be saved to. This is a required argument.
`start_date`	Integer vector of length 2, Start date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is start of series.

Details

Version 1.4, 2026-07-01

Value

File with time series will be produced.

Author(s)

Brian C. Monsell, `<monsell.brian@bls.gov>` or `<bcmonsell@gmail.com>`

References

Monsell BC, Lytras D, Findley D (2016). "Getting Started with X-13ARIMA-SEATS Input Files." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/gettingstartedx13-winx13.pdf.

Examples

```
UKgas_tc_date <- c(1970, 2)
UKgas_tc_1970_2 <-
   gen_tc_outlier_ts(UKgas_tc_date,
      this_start = c(1960, 1),
      this_end = c(1990, 4),
      this_freq = 4,
      tc_alpha = 0.9,
      return_matrix = TRUE)
```

```
## Not run: save_user_reg(UKgas_tc_1970_2, 'UKgas_tc_1970_2.txt')
## Not run: convert_dv_to_csv('UKgas_tc_1970_2.txt', 'UKgas_tc_1970_2.csv')
```

---

convert_identify_acf *Convert matrix of ACFs or PACFs generated by X-13ARIMA-SEATS* `identify` *spec to a list object*

Description

Generates a list of the ACF or PACF generated by the `identify` spec. For more information on the `identify` spec, see Section 7.9 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
convert_identify_acf(seas_obj = NULL, this_plot = "iac")
```

Arguments

seas_obj	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
this_plot	Character string; three character code for the type of plot to be generated. Allowed entries are "iac" (sample autocorrelation function, default), and "ipf" (sample partial autocorrelation function).

Details

Version 3.1, 2026-03-06

Value

A list of matrices of ACF or PACFs produced for different orders of differncing. The list entries are named based on the orders of differencing (`d0sd0` denotes no regular difference, no seasonal difference, `d1sd0` denotes one regular difference, no seasonal difference, etc.)

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
shoes_identify_seas <-
   seasonal::seas(shoes2008,
      identify.diff = c(0, 1),
      identify.sdiff = c(0, 1),
      identify.save = c("iac", "ipc"),
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      forecast.maxlead = 36,
      check.maxlag = 36,
      check.acflimit = 1.96,
      check.qlimit = 0.01,
      check.print = c( 'pacf', 'pacfplot' ))
shoes_identify_acf_list <-
   convert_identify_acf(shoes_identify_seas, 'iac')
shoes_identify_pacf_list <-
   convert_identify_acf(shoes_identify_seas, 'ipc')
```

`convert_x13save_to_csv` *convert a* `x13save` *formatted file into a csv file ready to be read by Python*

Description

Process a data file stored in `x13save` form and produce a `.csv` file that is ready to be read by Python. For more information on the `x13save` format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
convert_x13save_to_csv(
  this_x13save_file = NULL,
  this_csv_file = NULL,
  start_date = NULL
)
```

Arguments

`this_x13save_file`
: Character string; name of file with a time series stored in `x13save` format. This is a required argument.

`this_csv_file`
: Character string; name of file time series array to be saved to. This is a required argument.

`start_date`
: Integer vector of length 2, Start date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is start of series.

Details

Version 1.5, 2026-07-01

Value

File with time series will be produced.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Monsell BC, Lytras D, Findley D (2016). "Getting Started with X-13ARIMA-SEATS Input Files." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/gettingstartedx13-winx13.pdf.

Examples

```
## Not run: save_series(UKgas, 'UKgas.dat', save_dv = FALSE)
## Not run: convert_x13save_to_csv('UKgas.dat', 'UKgas.csv')
```

---

d11f_test — *D11 F-test for residual seasonality*

Description

Generates X-11's F-test for residual seasonality in the seasonally adjusted data. For more information on X-11 diagnostics, see Ladiray and Quenneville (2001) and Lothian and Morry (1978).

Usage

```
d11f_test(seas_obj = NULL, p_level = 0.01, return_this = "test")
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`p_level`	P-level used to test the D11 f-test for residual seasonality Default is `0.01`.
`return_this`	Character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 5.1, 2026-03-05

Value

A text string denoting if series passes or has a warning for residual seasonality. If D11f statistic not found, return `'none'`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ladiray D, Quenneville B (2001). *Seasonal Adjustment with the X-11 Method*, volume 158. Springer, New York. Lecture Notes in Statistics.

Lothian J, Morry M (1978). "A Test of Quality Control Statistics for the X-11-ARIMA Seasonal Adjustment Program." Research Paper, Seasonal Adjustment and Time Series Staff, Statistics Canada., https://www.census.gov/content/dam/Census/library/working-papers/1978/adrm/lothianmorry1978.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
                  series.period = 4,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=20,
                  check.print = c( "pacf", "pacfplot" ),
                  x11="")
ukgas_d11f_test <-
   d11f_test(ukgas_seas,
             p_level = 0.05,
             return_this = 'both')
```

`d11f_test_why` *D-11 F-test Warning Message*

Description

Shows why D11 F-test for residual seasonality fails. For more information on X-11 diagnostics, see Ladiray and Quenneville (2001) and Lothian and Morry (1978).

Usage

```
d11f_test_why(udg_list = NULL, p_level = 0.01, return_both = FALSE)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`p_level`	P-level used to test the d11 f-test for residual seasonality Default is `0.01`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 5.0, 2026-02-24

Value

A text string denoting why a series fails or has a warning for residual seasonality. If D11f statistic not found, return 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ladiray D, Quenneville B (2001). *Seasonal Adjustment with the X-11 Method*, volume 158. Springer, New York. Lecture Notes in Statistics.

Lothian J, Morry M (1978). "A Test of Quality Control Statistics for the X-11-ARIMA Seasonal Adjustment Program." Research Paper, Seasonal Adjustment and Time Series Staff, Statistics Canada., https://www.census.gov/content/dam/Census/library/working-papers/1978/adrm/lothianmorry1978.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas, series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      x11="", transform.function = "log",
      forecast.maxlead=20,
      check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_d11f_why <-
   d11f_test_why(ukgas_udg, p_level = 0.05, return_both = TRUE)
```

`difference_dates` *Date difference*

Description

Generate the difference between two dates. This function uses the `lubridate` package (Grolemund and Wickham 2011).

Usage

```
difference_dates(
  start_date = NULL,
  end_date = NULL,
  this_frequency = 12,
  include_start = TRUE,
  return_absolute = FALSE
)
```

Arguments

- `start_date` — Integer array of length 2, a date where the first element is the year and the second element is the month or quarter. This is a required entry.
- `end_date` — Integer array of length 2, a date to comapare to `this_date`. This is a required entry.
- `this_frequency` — Integer scalar, frequency of target time series. Entries limited to `1` (yearly), `2` (biannual), `3` (triannual), `4` (quarterly), `6` (Bimonthly), `365` (daily) and `12`. Default is `12` (monthly).
- `include_start` — Logical scalar; if `TRUE`, include the starting date in the difference, otherwise, the difference is left as computed. Default is `TRUE`.
- `return_absolute` — Logical scalar; if `TRUE`, returns the absolute value of the number of days if the difference is negative

Details

Version 4.0, 2026-02-24

Value

Integer scalar; difference between first date and second date.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). "Dates and Times Made Easy with lubridate." *Journal of Statistical Software*, **40**(3), 1–25. https://www.jstatsoft.org/v40/i03/.

Examples

```
diff_start <- difference_dates(c(1990,1), c(1995,1))
```

`employment_data_mts` *US Employment Series, four main components in an mts object*

Description

An `mts` object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
employment_data_mts
```

Format

An mts object with 4 time series elements in four columns:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

---

employment_list *US Employment Series, four main components in a list object*

Description

A list object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
employment_list
```

Format

A list object with 4 time series elements:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

---

fix_diag_list *Fix Diagnostic List*

Description

Fix an incomplete diagnostic list by filling in missing elements with NAs.

Usage

```
fix_diag_list(this_test = NULL, this_names = NULL, return_this = "both")
```

Arguments

this_test	list object of a seasonal adjustment or modeling diagnostic. This is a required entry.
this_names	character vector; complete set of names to check against. This is a required entry.
return_this	character string; what the function returns - 'test' returns test results, 'why' returns why the test failed or received a warning, or 'both'. Default is 'both'.

Details

Version 1.7, 2026-02-24

Value

diagnostic list object with missing names filled in.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
xt_composite_seas <- seasonal::seas(xt_data_new,
     transform.function = "log",
     check.print = c("none", "+acf", "+acfplot", "+normalitytest"),
     regression.aictest = NULL,
     outlier.types = "all",
     arima.model = "(0 1 1)(0 1 1)",
     list = list(
         list(x11 = ""),
         list(x11 = ""),
         list(seats.save = c("s11", "s12", "s10")),
         list(x11 = "")
     ),
)
xt_comp_update <-
     Filter(function(x) inherits(x, "seas"), xt_composite_seas)
xt_test_m7 <- lapply(xt_comp_update, function(x)
     try(mq_test(x, return_this = 'both')))
test_names <- names(xt_data_new)
num_names <- length(test_names)

if (!is.null(xt_test_m7)) {
    if (length(xt_test_m7) < num_names) {
      xt_test_m7 <-
         fix_diag_list(xt_test_m7, test_names, return_this = 'both')
    }
}
```

`gen_ao_outlier_ts` *Generate level change regression variable as a `ts` object*

Description

Generates a `ts` object for a AO (point) outlier regressor. For more information on AO outliers, see Sections 4.3 and 7.11 of U. S. Census Bureau (2025).

Usage

```
gen_ao_outlier_ts(
  ao_date = NULL,
  this_start = NULL,
  this_end = NULL,
  this_freq = 12,
  return_matrix = TRUE
)
```

Arguments

`ao_date`	Integer vector of length two - dates for AO outlier to be generated. This is a required entry.
`this_start`	Numeric vector; start date of AO outlier regressor generated. This is a required entry.
`this_end`	Numeric vector; end date of AO outlier regressor generated. This is a required entry.
`this_freq`	Numeric scalar; frequency of time series. Default: `12`, for a monthly series.
`return_matrix`	Logical scalar; If `TRUE`, the object returned is a one column time series matrix object. Default: `TRUE`.

Details

Version 2.0, 2026-02-24

Value

A `ts` object of a point outlier regressor.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
UKgas_ao_date <- c(1970, 2)
UKgas_ao_1970_2 <-
   gen_ao_outlier_ts(UKgas_ao_date,
                     this_start = c(1960, 1),
                     this_end = c(1990, 4),
                     this_freq = 4)
```

`gen_ar_spec_cv` *AR(30) Spectrum Critical Value*

Description

Generate critical values for AR(30) spectrum as in Maravall (2012). For more information on the AR(30) specrum, see Section 6.1.2 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
gen_ar_spec_cv(n_sim = 1e+05, series_length = 121, freq = 12)
```

Arguments

`n_sim`	Integer scalar; number of simulations; default is `100000`.
`series_length`	Integer scalar; length of each series simulated. Default is `121`.
`freq`	Integer scalar; frequency of the time series; default is `12` (monthly).

Details

Version 2.1, 2026-03-06

Value

List of critical values for each seasonal frequency for the 95th and 99th percentile.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ar30_spec_cv <- gen_ar_spec_cv(1000, 97, 12)
```

gen_hybrid_sa *Generate a hybrid seasonal adjustment*

Description

Generates a "hybrid" seasonal adjustment by replacing a span of a multiplicative seasonal adjustment with an additive adjustment. For more information on the need for hybrid seasonal' adjustment, see Monsell (2021).

Usage

```
gen_hybrid_sa(
  this_mult_sa = NULL,
  this_add_sa = NULL,
  this_start_hybrid = NULL,
  this_end_hybrid = NULL
)
```

Arguments

`this_mult_sa` Time series object of a multiplicative seasonal adjustment. This is a required entry.

`this_add_sa` Time series object of an additive seasonal adjustment. This is a required entry.

`this_start_hybrid` Integer vector of length 2, start of the span where additive adjustments replace multiplicative adjustment. This is a required entry.

`this_end_hybrid` Integer vector of length 2, end of the span where additive adjustments replace multiplicative adjustment. This is a required entry.

Details

Version 3.0, 2026-02-25

Value

Time series object with hybrid seasonal adjustment.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Monsell BC (2021). “Time Series Responses to the COVID-19 Pandemic at BLS for Monthly and Weekly Series.” https://www.bls.gov/osmr/research-papers/2021/pdf/st210040.pdf.

Examples

```
air_mult_seas <- seasonal::seas(AirPassengers, transform.function = "log")
air_mult_sa   <- seasonal::final(air_mult_seas)
air_add_seas  <- seasonal::seas(AirPassengers, transform.function = "none")
air_add_sa    <- seasonal::final(air_add_seas)
air_hybrid_sa <- gen_hybrid_sa(air_mult_sa, air_add_sa, c(1956,1), c(1956,12))
```

---

gen_ls_outlier_ts   *Level change regression variable as a* ts *object*

Description

Generates a ts object for a LS (level shift) outlier regressor. For more information on LS outliers, see Sections 4.3 and 7.11 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
gen_ls_outlier_ts(
  ls_date = NULL,
  this_start = NULL,
  this_end = NULL,
  this_freq = 12,
  x13type = TRUE,
  return_matrix = TRUE
)
```

Arguments

ls_date	Integer vector of length two - dates for LS outlier to be generated. This is a required entry.
this_start	Numeric vector; start date of LS outlier regressor generated. This is a required entry.
this_end	Numeric vector; end date of LS outlier regressor generated. This is a required entry.
this_freq	Numeric scalar; frequency of time series. Default: 12, for a monthly series
x13type	Logical scalar; Indicates if level change outlier is defined as in X-13ARIMA-SEATS. Otherwise, the regressor is defined as 0 before the change in level, and 1 otherwise. Default: TRUE
return_matrix	Logical scalar; If true, the object returned is a one column time series matrix object. Default: TRUE

Details

Version 2.0, 2026-02-24

Value

Generate ts object of a level change outlier regressor

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
UKgas_ls_date <- c(1970, 2)
UKgas_ls_1970_2 <-
   gen_ls_outlier_ts(UKgas_ls_date,
                     this_start = c(1960, 1),
                     this_end = c(1990, 4),
                     this_freq = 4)
```

---

gen_tc_outlier_ts *Temporary change outlier regression as a* ts *object*

Description

Generates a ts object for a TC (temporary change) outlier regressor For more information on TC outliers, see Sections 4.3 and 7.11 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
gen_tc_outlier_ts(
  tc_date = NULL,
  this_start = NULL,
  this_end = NULL,
  this_freq = 12,
  tc_alpha = NULL,
  return_matrix = TRUE
)
```

Arguments

tc_date	Integer vector of length two - dates for TC outlier to be generated. This is a required entry.
this_start	Numeric vector; start date of TC outlier regressor generated. This is a required entry.
this_end	Numeric vector; end date of TC outlier regressor generated. This is a required entry.
this_freq	Numeric scalar; frequency of time series. Default: 12, for a monthly series
tc_alpha	Numeric scalar; Rate of decay for the TC outlier. Default: will be computed as in X-13ARIMA-SEATS for a weekly series; see section 7.11 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025) for more details.
return_matrix	Logical scalar; If true, the object returned is a one column time series matrix object. Default: TRUE.

Details

Version 2.0, 2026-02-24

Value

`ts` object for a temporary change outlier regressor.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
UKgas_tc_date <- c(1970, 2)
UKgas_tc_1970_2 <-
   gen_tc_outlier_ts(UKgas_tc_date, this_start = c(1960, 1), this_end = c(1990, 4),
                     this_freq = 4, tc_alpha = 0.9, return_matrix = TRUE)
```

---

gen_x13save_header — *Generate header for* x113save *formatted file*

Description

Generate the header at the top of an X-13ARIMA-SEATS' `x13save` formatted file. For more information on the `datevalue` format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
gen_x13save_header(this_series)
```

Arguments

`this_series`	Time series array to be saved. This is a required entry.

Details

Version 1.1, 2026-07-01

Value

Header for X-13ARIMA-SEATS `x13save` formatted file. Name taken for second column is taken from `this_series`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_header <- gen_x13save_header(UKgas)
```

`gen_x13_table_list` *X-13 Tables Available*

Description

Generates a list of X-13 tables that can be extracted with the `seasonal` package (Sax and Eddelbuettel 2018).

Usage

```
gen_x13_table_list(this_table_type = "all", what_code = "both")
```

Arguments

`this_table_type`
: Vector of character strings listing types of X-13 tables to output. Default is `'all'`; other choices are `'diagnostics'`, `'matrices'`, `'spectrum'`, `'timeseries'`.

`what_code`
: Character scalar; which codes are returned. Default is `'all'`; other choices are `'short'` (for short codes), or `'long'` (for the longer code names).

Details

Version 4.0, 2026-06-15

Value

A list of arrays with table names and abbreviations from X-13ARIMA-SEATS in several different elements specified by the user: `diagnostics`, `matrices`, `spectrum`, `timeseries`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
x13_tables_all <- gen_x13_table_list()
x13_tables_spectrum_short <-
    gen_x13_table_list(this_table_type = "spectrum", what_code = "short")
```

get_acf *Get ACF values, diagnostics*

Description

Extract values from the sample ACF of the regARIMA model residuals, along with Ljung-Box Q values (Ljung and Box 1978).

Usage

```
get_acf(
  seas_obj = NULL,
  lags = NULL,
  as_data_frame = TRUE,
  add_rownames = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
lags	An array of integer lags to extract from the ACF output. If not specified, all lags will be returned.
as_data_frame	Logical scalar. If `TRUE`, the values are returned as a data frame. If `FALSE`, the result is returned as a matrix object. Default is `TRUE`.
add_rownames	Logical scalar. If `TRUE`, row names are constructed from the lags specified by the user. If `FALSE`, no rownames will be added. Default is `TRUE`.

Details

Version 3.1, 2026-03-06

Value

A data frame (or matrix) with 5 columns: `"Sample_ACF"` (sample ACF Value), `"SE_of_ACF"` (ACF standard error), `"Ljung.Box_Q"` (Ljung-Box Q statistic), `"df_of_Q"` (degrees of freedom for Ljung=Box Q), and `"P.value"` (p-value for Ljung Box Q).

Author(s)

Brian C. Monsell, <`monsell.brian@bls.gov`> or <`bcmonsell@gmail.com`>

References

Ljung GM, Box GEP (1978). “On a Measure of Lack of Fit in Time Series Models.” *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      arima.model="(0 1 1)(0 1 1)",
      check.save = "acf",
      check.maxlag = 36,
      forecast.maxlead = 36, slidingspans = "",
      transform.function = "log")
air_acf_df <-
   get_acf(air_seas, lags = c(1,2,3,12,13,24))
```

---

get_arima_estimates_matrix

ARMA Coefficient Summary

Description

Generate a summary of ARMA coefficients for a single series.

Usage

```
get_arima_estimates_matrix(seas_obj = NULL, add_diff = FALSE)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
add_diff	logical scalar; add differencing information, if included in the model.

Details

Version 4.0, 2026-02-25

Value

matrix of ARMA coefficients, standard errors, and t-statistics for a given series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  regression.aictest = 'td',
                  check.print = c( "pacf", "pacfplot" ),
                  forecast.maxlead=36,
                  x11="",
                  slidingspans = "")
air_arima_matrix <-
   get_arima_estimates_matrix(air_seas, add_diff = TRUE)
```

---

get_auto_outlier_string

Get automatic outlier names

Description

Get the names of outliers identified in the seas object for a single series. Details on outliers (Sections 4.3 and 7.13) and outlier identification (Sections 4.7 and 7.11) can be found in the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
get_auto_outlier_string(seas_obj = NULL)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

Details

Version 4.1, 2026-03-05

Value

Character string containing a summary of the outliers identified in the regARIMA model. If no regressors or automatic outliers in the model, the routine will return a blank character.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="")
this_auto_outlier <- get_auto_outlier_string(air_seas)
```

---

get_fcst_tval *T-values of within sample forecasts*

Description

Returns t-values of within sample forecasts, up to 3.

Usage

```
get_fcst_tval(seas_obj = NULL, terror_lags = NULL)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
terror_lags	Integer scalar for number of forecast lags from the end of series where we'll collect t-statistics. Must be either 1, 2, or 3. This is a required entry.

Details

Version 3.1, 2026-04-14

Value

A numeric array of t-values of within sample forecasts, up to length 3.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_short <-
   seasonal::seas(AirPassengers,
                  series.span = ',1960.9',
                  transform.function = "log",
                  arima.model="(0 1 1)(0 1 1)",
                  forecast.maxlead = 36,
                  x11 = "",
                  slidingspans = "")
fcst_tstat <- get_fcst_tval(air_seas_short, 3)
```

get_ftest_from_udg *Get F-test info from external UDG file*

Description

Parse external udg file and get information for model-based F-test for a regressor.

Usage

```
get_ftest_from_udg(this_base = NULL, this_path = NULL, this_reg = "Seasonal")
```

Arguments

`this_base`	Character scalar; Filename of output file of the X-13 run. This is a required entry.
`this_path`	Character scalar; Path of udg file. This is an optional entry.
`this_reg`	Character scalar; Type of regressor for model-based F-test. Default is `Seasonal`.

Details

Version 1.2, 2026-02-25

Value

Numeric vector of the degrees of freedom, F-test, and p-value for this model based F-test.

Author(s)

Brian C. Monsell, <`monsell.brian@bls.gov`> or <`bcmonsell@gmail.com`>

Examples

```
## Not run: this_ftest<-
             get_ftest_from_udg(this_file = "AE1011330000_auto",
                                this_path = "X:/code/census_build_60/basic/",
                                this_reg = "Trigonometric Seasonal")
## End(Not run)
```

get_model_ftest *Get model based F-test*

Description

Extract values associated with the model based F-test specified by the `this_ftest` argument.

Usage

```
get_model_ftest(seas_obj = NULL, this_ftest = "seasonal", return_this = "all")
```

Arguments

- `seas_obj` A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
- `this_ftest` Character string; type of model based f-test to return. Default is `"seasonal"`; only other acceptable value is `"td"`.
- `return_this` Character string, Code that controls what values are returned. Acceptable values are `"all"`, `"dof"` (degrees of freedom), `"ftest"`(F-test value), or `"pval"` (F-test p-value). Default is `"all"`.

Details

Version 3.8, 5/15/2024

Value

Numeric vector with seasonal or trading day F-statistic, degrees of freedom, or p-value, depending on the value of `return_this`. If not found, return `NULL`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_td_seasonal <-
    seasonal::seas(AirPassengers, arima.model = "(0 1 1)",
                   regression.variables = c("seasonal", "td"), x11="")
this_seas_ftest <-
   get_model_ftest(air_seas_td_seasonal, this_ftest = "seasonal",
                   return_this = "ftest")
this_td_ftest   <-
   get_model_ftest(air_seas_td_seasonal, this_ftest = "td",
                   return_this = "ftest")
```

`get_month_index` *Generate index of month abbreviation*

Description

Process string of month abbreviation to return a numeric index.

Usage

```
get_month_index(this_month_string, full_name = FALSE)
```

Arguments

this_month_string
: Character string; 3 character abbreviation of month. This entry is required.

full_name
: Logical scalar; indicates if this is the full name of the month.

Details

Version 3.0, 2026-06-15

Value

Index of month - 1 for 'Jan', 2 for 'Feb', etc.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
thisOtl <- 'AO2015.Jan'
thisCode <- 'AO'
thisPerChar <- substr(thisOtl,nchar(thisCode)+6,nchar(thisOtl))
thisPerIndex <- get_month_index(thisPerChar)
```

---

get_mq_key — *Make a UDG key for X-11-ARIMA M and Q statistics*

Description

Generates the UDG key for X-11-ARIMA M and Q statistics based on a label. For more information on the X-11-ARIMA quality control statistics, see Lothian and Morry (1978) and Ladiray and Quenneville (2001).

Usage

```
get_mq_key(this_label = NULL)
```

Arguments

this_label
: Character string; name of an X-11-ARIMA M and Q statistics. This is a required entry.

Details

Version 2.0, 2026-03-06

Value

Character string with the corresponding UDG label for this_label. If incorrect label is specified, returns NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ladiray D, Quenneville B (2001). *Seasonal Adjustment with the X-11 Method*, volume 158. Springer, New York. Lecture Notes in Statistics.

Lothian J, Morry M (1978). "A Test of Quality Control Statistics for the X-11-ARIMA Seasonal Adjustment Program." Research Paper, Seasonal Adjustment and Time Series Staff, Statistics Canada., https://www.census.gov/content/dam/Census/library/working-papers/1978/adrm/lothianmorry1978.pdf.

Examples

```
m7_key  <- get_mq_key('M7')
```

`get_mq_label` *Make a label for X-11-ARIMA M and Q statistics*

Description

Generates a label for X-11-ARIMA M and Q statistics For more information on the X-11-ARIMA quality control statistics, see Lothian and Morry (1978) and Ladiray and Quenneville (2001).

Usage

```
get_mq_label(this_key = "f3.q")
```

Arguments

this_key	Character string; name of an X-11-ARIMA M and Q statistics used in the UDG X-13 output. Default is "f3.q".

Details

Version 2.0, 2026-03-06

Value

Character string with the corresponding label for `this_key`. If incorrect label is specified, returns `NULL`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ladiray D, Quenneville B (2001). *Seasonal Adjustment with the X-11 Method*, volume 158. Springer, New York. Lecture Notes in Statistics.

Lothian J, Morry M (1978). "A Test of Quality Control Statistics for the X-11-ARIMA Seasonal Adjustment Program." Research Paper, Seasonal Adjustment and Time Series Staff, Statistics Canada., https://www.census.gov/content/dam/Census/library/working-papers/1978/adrm/lothianmorry1978.pdf.

Examples

```
m7_label  <- get_mq_label('f3.m07')
```

---

get_nonseasonal_theta   *Nonseasonal Moving Average from Airline Model*

Description

Get the value of a nonseasonal moving average coefficient estimated from an airline model.

Usage

```
get_nonseasonal_theta(
  seas_obj = NULL,
  this_index = 1,
  return_string = TRUE,
  significant_digits = 3
)
```

Arguments

- `seas_obj` — A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
- `this_index` — An integer scalar, an index of the vector values to be passed. Acceptable values are `1` (nonseasonal MA coefficient value), `2` (nonseasonal MA coefficent standard error), or `3` (t-value of the nonseasonal MA coefficient). Default is `1`.
- `return_string` — A Logical scalar; indicates whether value returned is a string or numeric. Default is `TRUE`.
- `significant_digits` — An integer scalar; significant digits to be saved when a string is returned. Default is `3`.

Details

Version 2.0, 2026-02-25

Value

Character string containing a value related to the seasonal MA coefficient from the regARIMA model fit in the `seas` object `seas_obj`. If `return_string` is `FALSE`, this is a numeric. The standard error or t-value of the seasonal MA coefficient can be returned depending on the value of `this_index`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  x11="")
this_nonseasonal_theta <-
   get_nonseasonal_theta(air_seas, return_string = FALSE)
```

---

get_norm_stat *Extract normality statistics from X-13*

Description

Extract normality statistics from the seas object of a single series.

Usage

```
get_norm_stat(seas_obj = NULL, this_norm = NULL)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_norm	Character string; type of normality statistic being extracted. Permissable values are 'a', 'kurtosis', 'skewness'. This is a required entry.

Details

Version 4.1, 2026-03-05

Value

Double precision number for normality statistic described in this_key. If incorrect this_key used, function returns a NULL value. If normality statistic not generated in this run, function returns a NULL value. Descriptions of these diagnostics can be found in Section 7.3 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "",
                  slidingspans = "")
air_norm_stat <-
   get_norm_stat(air_seas, this_norm = 'kurtosis')
```

`get_outlier_list` *List of outliers generated from a list of seasonal objects*

Description

Generate a summary of regARIMA model outliers from a list of seas objects. For more information on outliers, see Table 4.1 of Section 4.3 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025). Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
get_outlier_list(seas_obj_list = NULL)
```

Arguments

`seas_obj_list`	List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.

Details

Version 3.1, 2026-03-06

Value

vector of regARIMA model outliers for a given series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
      transform.function = "log",
      outlier.types = "all",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead=36, x11 = "",
      check.print = c( "pacf", "pacfplot" ))
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_diag <-
    get_outlier_list(unemp_seas_update)
```

---

get_regarima_estimates_matrix

Generate summary of regARIMA model coefficients

Description

Generate a summary of coefficients from a regARIMA model for a single series.

Usage

```
get_regarima_estimates_matrix(
  seas_obj = NULL,
  add_diff = FALSE,
  this_xreg_names = NULL
)
```

Arguments

`seas_obj` A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.

`add_diff` logical scalar; add differencing information, if included in the model.

`this_xreg_names` Character array; name of user defined regressors. Default is `NULL`, no user defined regressors. Number of names in this vector should match number of user-defined regressors; if not, a warning message will be produced.

Details

Version 3.0, 2026-02-25

Value

A matrix of regARIMA model coefficients, standard errors, and t-statistics for a given series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36,
                  regression.aictest = 'td',
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "", slidingspans = "")
air_regarima_matrix <- get_regarima_estimates_matrix(air_seas)
```

---

get_regression_code *Get X-13 entries for regression spec*

Description

Get potential input for X-13ARIMA-SEATS for the `regression` spec for a given series.

Usage

```
get_regression_code(seas_obj = NULL, this_argument = "variables")
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_argument`	Character string; argument from the X-13ARIMA-SEATS `regression` spec. Possible values are `"variables"`, `"user"`, `"start"`, `"data"`, `"usertype"`, or `"b"`. Default is `"variables"`.

Details

Version 1.1, 2026-04-22

Value

Vector containing the entry for a `regression` argument from `seas_obj`. If there is no entry for that argument, the routine will return a `NULL` object.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "", slidingspans = "")
air_reg_var <- get_regression_code(air_seas)
air_reg_b   <- get_regression_code(air_seas, "b")
```

`get_regression_estimates_matrix`
Generate regression coefficient summary

Description

Generate a summary of regression coefficients for a single series.

Usage

```
get_regression_estimates_matrix(seas_obj = NULL, this_xreg_names = NULL)
```

Arguments

`seas_obj` A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.

`this_xreg_names` Character array; name of user defined regressors. Default is `NULL`, no user defined regressors.

Details

Version 3.4, 2024-05-14

Value

Matrix of regression coefficients, standard errors, and t-statistics for a given series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36,
                  regression.aictest = 'td',
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "", slidingspans = "")
air_reg_matrix <- get_regression_estimates_matrix(air_seas)
```

`get_reg_string` *Get names of regressors*

Description

Generate string of names for the regressors used in the model fit for a given series.

Usage

```
get_reg_string(seas_obj = NULL, xreg_names = NULL, exclude_user = FALSE)
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`xreg_names`	Character vector with names of user defined regressors used in model. Default is `NULL`, no user defined regressors. Number of names in this vector should match number of user-defined regressors; if not, a warning message will be produced.
`exclude_user`	Logical scalar; determines if user defined regressors are included in the string of regression names output by the function. Default is `FALSE`.

Details

Version 4.0, 2026-02-25

Value

Character string containing a summary of the regressors in the regARIMA model. If no regressors in the model, the routine will return a blank character.

Author(s)

Brian C. Monsell, <`monsell.brian@bls.gov`> or <`bcmonsell@gmail.com`>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "", slidingspans = "")
air_reg <- get_reg_string(air_seas)
```

---

get_seasonal_ftest_all *Generate model based F-test*

Description

Generate model based F-test, changing the model to remove seasonal differences and adding seasonal regressors if necessary. This function is used in the overall seasonal test from Maravall (2012). For more information on the model based seasonal F-test with applications, see Bell et al. (2022) and Monsell (2024).

Usage

```
get_seasonal_ftest_all(
  seas_obj = NULL,
  this_series = "b1",
  use_seasonal = TRUE
)
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_series`	Character string; the table used to generate the model based F-test. Default is `"b1"`.
`use_seasonal`	Logical scalar; if TRUE, the `seasonal` regressor is used; otherwise, use the sine-cosine trignonmetric regressors generated by `sincos`.

Details

Version 5.2, 2026-04-16

Value

A numeric vector with the degrees of freedom, F statistic, and probability generated for the model based seasonal f-test used in the seasonal testing procedure in Maravall (2012).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Monsell BC (2024). "Two Topics in Applying Seasonal Adjustment Diagnostics on Large Sets of Series." https://www.bls.gov/osmr/research-papers/2024/pdf/st240110.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  series.save = "a1",
                  transform.function = "log",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "", slidingspans = "")
air_ftest_all <-
    get_seasonal_ftest_all(air_seas, this_series = "a1")
```

---

get_seasonal_ftest_prob

Probability of model based F-test

Description

Get probability for model based F-test, changing the model to remove seasonal differences and adding seasonal regressors if necessary. This function is used in the overall seasonal test from Maravall (2012). For more information on the model based seasonal F-test with applications, see Bell et al. (2022) and Monsell (2024).

Usage

```
get_seasonal_ftest_prob(
  seas_obj = NULL,
  this_series = "b1",
  use_seasonal = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; the table used to generate the model based F-test. Default is "b1".
use_seasonal	Logical scalar; if TRUE, the seasonal regressor is used; otherwise, use the sine-cosine trignonmetric regressors generated by sincos.

Details

Version 5.3, 2026-04-16

Value

Test probability generated for the model based seasonal F-test used in the seasonal testing procedure in Maravall (2012).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Monsell BC (2024). "Two Topics in Applying Seasonal Adjustment Diagnostics on Large Sets of Series." https://www.bls.gov/osmr/research-papers/2024/pdf/st240110.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_seasonal <-
  seasonal::seas(AirPassengers,
                 series.save = "a1",
                 transform.function = "log",
                 arima.model = "(0 1 1)",
                 regression.variables = "seasonal",
                 forecast.maxlead = 36,
                 slidingspans = "")
air_ftest_prob <-
   sautilities::get_seasonal_ftest_prob(air_seas_seasonal)
```

get_seasonal_theta *Seasonal Moving Average from Airline Model*

Description

Get the value of a seasonal moving average coefficient estimated from an airline model.

Usage

```
get_seasonal_theta(
  seas_obj = NULL,
  freq = 12,
  this_index = 1,
  return_string = TRUE,
  significant_digits = 3
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
freq	A numeric scalar, the frequency of the time series. Default is 12.
this_index	An integer scalar, an index of the vector values to be passed. Acceptable values are 1 (seasonal MA coefficient value), 2 (seasonal MA coefficent standard error), or 3 (t-value of the Seasonal MA coefficient). Default is 1.
return_string	A Logical scalar; indicates whether value returned is a string or numeric. Default is TRUE.
significant_digits	An integer scalar; significant digits to be saved when a string is returned. Default is 3.

Details

Version 3.0, 2026-02-25

Value

Character string containing a value related to the seasonal MA coefficient from the regARIMA model fit in the seas object seas_obj. The standard error or t-value of the seasonal MA coefficient can be returned depending on the value of this_index. If return_string is FALSE, this is a numeric.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  arima.model = "(0 1 1)(0 1 1)",
                  x11="")
this_seasonal_theta <-
    get_seasonal_theta(air_seas, return_string = FALSE)
```

`get_timer` *Generate elapsed time of run in miliseconds*

Description

Read profiler information from saved files and generate the elapsed time of an X-13 run in miliseconds.

Usage

```
get_timer(this_base = NULL, this_path = NULL)
```

Arguments

`this_base`	Character scalar; Filename of output file of the X-13 run. This is a required entry.
`this_path`	Character scalar; Path of profiler output. This is an optional entry.

Details

Version 1.1, 2023-11-29

Value

Numeric object of the elapsed time of the X-13 run.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
## Not run: AE1011330000_time <-
            get_timer("AE1011330000_auto", "X:/code/census_build_60/basic/")
## End(Not run)
```

get_transform *Get transformation*

Description

Get transformation from the seas object of a single time series.

Usage

```
get_transform(seas_obj = NULL, return_code = TRUE)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
return_code	logical scalar; if TRUE, return input for X-13. Default is TRUE.

Details

Version 3.1, 2026-04-21

Value

Character string with transformation used to model time series in seas run.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
                  arima.model = "(0 1 1)(0 1 1)",
                  x11="")
air_trans <- get_transform(air_seas)
air_trans_label <-
   get_transform(air_seas, return_code = FALSE)
```

get_udg_entry *Get entry from UDG file*

Description

Returns a specific element of a list of udg entries.

Usage

```
get_udg_entry(
  seas_obj = NULL,
  this_key = NULL,
  this_index = 0,
  return_numeric = TRUE
)
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_key`	Character scalar; keyword found in UDG output generated by X-13ARIMA-SEATS. This is a required entry.
`this_index`	Integer scalar; index of entry in vector to extract. If set to 0 (the default), get the last entry. If less than zero, return the entire entry.
`return_numeric`	Logical scalar; if `TRUE`, convert character to numeric object. Default is `TRUE`.

Details

Version 3.1 2026-04-21

Value

The `this_index` element of the array from the UDG entry for `this_key`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_short <-
   seasonal::seas(AirPassengers,
                  series.span = ',1960.9',
                  transform.function = "log",
                  arima.model="(0 1 1)(0 1 1)",
                  forecast.maxlead = 36,
                  x11 = "",
```

```
                  slidingspans = "")
fcst_tstat <-
    get_udg_entry(air_seas_short, 'forctval01')
```

---

get_udg_index — *Index for entry in UDG list*

Description

Return index for entry in UDG list returned by the udg() function of the seasonal package.

Usage

```
get_udg_index(udg_list = NULL, this_key = NULL)
```

Arguments

udg_list	List object generated by udg function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_key	Keyword found in udg files generated by X-13ARIMA-SEATS. This is a required entry.

Details

Version 3.1, 2026-03-05

Value

An integer denoting which element in the udg output matches the key provided by the user. If there is no match, the function returns the number 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_udg_index_a <-
   get_udg_index(ukgas_udg, this_key='a')
```

get_udg_list — *Generate a list of* udg *elements from a list of* seas *objects.*

Description

Process a list of seas elements to get a list of results from the udg function of the seasonal package. Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
get_udg_list(seas_obj_list = NULL, this_key = "autoout")
```

Arguments

seas_obj_list	List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.
this_key	Character string containing keyword of the udg function that returns a numeric value. Default is 'autoout'.

Details

Version 2.2, 2026-04-22

Value

A list containing the results of extracting this_key from the udg output within the elements of seas_obj_list. An error message will be printed if try errors are found.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C (2024). “Adjusting Multiple Series — cran.r-project.org.” [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

Examples

```
emp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36, x11 = "",
                  check.print = c( "pacf", "pacfplot" ))
emp_seas_update <-
   Filter(function(x) inherits(x, "seas"), emp_seas_list)
xt_m7 <-
  get_udg_list(emp_seas_update, this_key = 'f3.m07')
xt_q2 <-
  get_udg_list(emp_seas_update, this_key = 'f3.qm2')
```

get_value_from_udg *Get value from external UDG file*

Description

Parse external udg file and get value for a user supplied key

Usage

```
get_value_from_udg(
  this_base = NULL,
  this_path = NULL,
  this_key = NULL,
  return_numeric = TRUE
)
```

Arguments

this_base	Character scalar; Filename of output file of the X-13 run. This is a required entry.
this_path	Character scalar; Path of profiler output. This is an optional entry.
this_key	Character scalar; Key from udg file to search for. This is a required entry.
return_numeric	Logical scalar; Value returned is converted to numeric. Default is TRUE, returns value as a number.

Details

Version 2.1, 2026-06-16

Value

Numeric value associated with this_key.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
## Not run: this_ftest <-
             get_ftest_from_udg(this_base = "AE1011330000_auto",
                                this_path = "X:/code/census_build_60/basic/",
                                this_key = "aicc")
## End(Not run)
```

get_window *Subspan time series*

Description

Generate subspan of time series.

Usage

```
get_window(X = NULL, plot_start = NULL, plot_end = NULL)
```

Arguments

X	Time Series (ts) object. This is a required entry.
plot_start	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
plot_end	Integer vector of length 2; Starting date for plot. Default is ending date for the time series.

Details

Version 2.3, 2023-05-25

Value

Generate subspan of time series X specified by plot_start and plot_end.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air50 <-
   get_window(AirPassengers, plot_start = c(1950,1), plot_end = c(1959,12))
```

get_x11_seasonal_ma *Get X-ll seasonal moving average*

Description

Returns the seasonal moving average used in an X-ll seasonal adjustment from a seas object.

Usage

```
get_x11_seasonal_ma(seas_obj = NULL)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

Details

Version 2.0, 2026-02-25

Value

Returns the seasonal moving average used in an X-ll seasonal adjustment from a seas object. Returns NULL if X-11 seasonal adjustment is not done.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_short <-
   seasonal::seas(AirPassengers,
                  series.span = ',1960.9',
                  transform.function = "log",
                  arima.model="(0 1 1)(0 1 1)",
                  forecast.maxlead = 36,
                  x11 = "",
                  slidingspans = "")
air_sf_short  <- get_x11_seasonal_ma(air_seas_short)
```

---

import_est   *Import regCMPNT estimates file*

Description

Reads in an estimated component from a file saved by regCMPNT (Bell 2011).

Usage

```
import_est(
  file_name = NULL,
  column_name = c("Unscaled_Stochastic", "Scale_Factors", "Scaled_Stochastic",
    "Regression_Effects", "Combined_Estimate"),
  return_matrix = TRUE
)
```

Arguments

`file_name`	Character string; file name for regCMPNT estimate file. This is a required entry
`column_name`	Array of character strings; names for the columns of the estimates matrix. Array must be of length 5. Default is `c("Unscaled_Stochastic", "Scale_Factors", "Scaled_Stochastic","Regression_Effects", "Combined_Estimate")`.
`return_matrix`	Logical scalar; determines if a matrix or data frame object is returned. Default is `TRUE`, a `ts` matrix object will be returned.

Details

Version 2.0, 2026-02-25

Value

A ts matrix object or a data frame of ts objects which contains the contents of the estimates for a given component from a regCMPNT run. The file name for the for the component file has an .est file extension.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell WR (2011). "RegCMPNT: A Fortran Program for Regression Models with ARIMA Components." *Journal of Statistical Software*, **41**(7), 1-23. https://doi.org/10.18637/jss.v041.i07.

Examples

```
## Not run:
n3000019_trend_df         <- import_est("n300019_rev2_comp01.est", return_matrix = FALSE)
n3000019_seasonal_df      <- import_est("n300019_rev2_comp02.est", return_matrix = FALSE)
n3000019_irregular_df     <- import_est("n300019_rev2_comp03.est", return_matrix = FALSE)
n3000019_samplingerror_df <- import_est("n300019_rev2_comp04.est", return_matrix = FALSE)

## End(Not run)
```

---

import_var *Import regCMPNT Variance file*

Description

Reads in variances for a component from a file saved by regCMPNT (Bell 2011).

Usage

```
import_var(file_name = NULL, column_name = NULL, return_matrix = TRUE)
```

Arguments

file_name	Character string; file name for regCMPNT variance file. This is a required entry
column_name	Array of character strings; names for the columns of the estimates matrix. Array must be of length 3 or 4. If regressors are used, the default is c("Unscaled_Stochastic", "Scaled_Stochastic", "Regression_Estimation", "Combined"). If regressors are not used, the default is c("Unscaled_Stochastic", "Scaled_Stochastic", "Combined").
return_matrix	Logical scalar; determines if a matrix object is returned. Default is TRUE, which forces the function to return a data frame object.

Details

Version 2.0, 2026-02-25

Value

A `ts` matrix object or a data frame of `ts` objects which contains the contents of the variances for a given component from a regCMPNT run. The file name for the for the component file has an `.var` file extension.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell WR (2011). "RegCMPNT: A Fortran Program for Regression Models with ARIMA Components." *Journal of Statistical Software*, **41**(7), 1-23. https://doi.org/10.18637/jss.v041.i07.

Examples

```
## Not run:
n3000019_trend_var_df      <- import_var("n300019_rev2_comp01.var", return_matrix = FALSE)
n3000019_seasonal_var_df    <- import_var("n300019_rev2_comp02.var", return_matrix = FALSE)
n3000019_irregular_var_df    <- import_var("n300019_rev2_comp03.var", return_matrix = FALSE)
n3000019_samplingerror_var_df <- import_var("n300019_rev2_comp04.var", return_matrix = FALSE)

## End(Not run)
```

`input_saved_x13_file` *Import File Saved by X-13ARIMA-SEATS*

Description

Import data from a file saved by the X-13ARIMA-SEATS program.

Usage

```
input_saved_x13_file(filename = NULL, pos = 2, ncol = 2)
```

Arguments

`filename`	Character string, filename of a file saved by the X-13ARIMA-SEATS program. This is a required entry.
`pos`	Integer scalar, column of data to be extracted from `filename`. Default is 2.
`ncol`	Integer scalar, number of columns of data that exist within `filename`. Default is 2.

Details

Version 1.2, 2023-05-25

Value

A time series object.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
## Not run:
airline.sa <- input_saved_x13_file("airline.d11")

## End(Not run)
```

---

lbq_fail *Ljung Box Q Test failure message*

Description

Tests whether the sample autocorrelation of the residuals from a time series model fails the Ljung-Box Q test (Ljung and Box 1978).

Usage

```
lbq_fail(this_acf = NULL, lbq_lags_fail, p_limit = 0.01)
```

Arguments

this_acf	Matrix object of the saved acf table. This is a required entry.
lbq_lags_fail	Lags of the ACF to test. Default is c(12, 24).
p_limit	Numeric limit for the p-value of the Ljung-Box Q. Default is 0.01.

Details

Version 2.3, 2026-04-16

Value

Logical object which is TRUE if series fails the Ljung Box Q test, FALSE otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM, Box GEP (1978). "On a Measure of Lack of Fit in Time Series Models." *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      x11="",
      transform.function = "log",
      forecast.maxlead=20,
      check.save = 'acf',
      check.print = c( "acf", "pacf", "pacfplot" ))
ukgas_acf <- seasonal::series(ukgas_seas, "acf")
ukgas_lbq_fail <-
   lbq_fail(ukgas_acf, lbq_lags_fail = c(4, 8))
```

---

lbq_fail_why *Produce an Ljung-Box Q test failure message.*

Description

Generates text on why the sample autocorrelation of the residuals from a time series model fails the Ljung-Box Q test (Ljung and Box 1978).

Usage

```
lbq_fail_why(
  this_acf = NULL,
  lbq_lags_fail = c(12, 24),
  p_limit = 0.01,
  return_both = FALSE
)
```

Arguments

this_acf	Matrix object of the saved acf table. This is a required entry.
lbq_lags_fail	Lags of the ACF to test. Default is `c(12, 24)`.
p_limit	Numeric limit for the p-value of the Ljung-Box Q. Default is `0.01`.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or just produce a warning. Default is `FALSE`.

Details

Version 2.3, 2026-04-16

Value

Character object tells why series fails the Ljung-Box Q test, `'pass'` otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM, Box GEP (1978). “On a Measure of Lack of Fit in Time Series Models.” *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      x11="",
      transform.function = "log",
      forecast.maxlead=20,
      check.save = 'acf',
      check.print = c( "acf", "pacf", "pacfplot" ))
ukgas_acf <- seasonal::series(ukgas_seas, "acf")
ukgas_lbq_fail_why <-
   lbq_fail_why(ukgas_acf, lbq_lags_fail = c(4, 8),
                return_both = TRUE)
```

`lbq_test` *Ljung-Box Q ACF test*

Description

Tests whether the residuals from a time series model has acceptable autocorrelation as indicated by the Ljung-Box Q test (Ljung and Box 1978).

Usage

```
lbq_test(
  seas_obj = NULL,
  lbq_lags_fail = c(12, 24),
  p_limit = 0.01,
  return_this = "test"
)
```

Arguments

`seas_obj`	A seas object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`lbq_lags_fail`	Integer vectorl; lags of the ACF to test. Default is `c(12, 24)`.
`p_limit`	Integer scalar; numeric limit for the p-value of the Ljung-Box Q. Default is `0.01`.
`return_this`	character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 2.2, 2026-03-09

Value

A text string denoting if series passes, fails, or has a warning for residual autocorrelation. If Ljung-Box not found, the seasonal object will be updated with check spec arguments - if this is unsuccessful, return 'none'. If regARIMA model not estimated, return 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM, Box GEP (1978). "On a Measure of Lack of Fit in Time Series Models." *Biometrika*, **65**, 297–304. doi:10.1093/biomet/65.2.297.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      check.save = 'acf',
      check.maxlag = 12,
      check.qlimit = 0.01,
      forecast.maxlead=20,
      x11="")
ukgas_lbq_test <-
   lbq_test(ukgas_seas,
      lbq_lags_fail = c(4, 8),
      return_this = 'both')
```

`make_diag_df` *Generate diagnostic summary data frame*

Description

Generates a data frame with seasonal adjustment and modeling diagnostics. For information on these diagnostics, see various sections of U. S. Census Bureau (2025) and Bell et al. (2022).

Usage

```
make_diag_df(
  this_data_names = NULL,
  this_acf_test = NULL,
  this_d11f_test = NULL,
  this_spec_peak_test = NULL,
  this_spec_peak_ori_test = NULL,
  this_qs_test = NULL,
  this_qs_rsd_test = NULL,
```

```
  this_qs_seasonal_test = NULL,
  this_model_test = NULL,
  this_sspan_test = NULL,
  this_m7_test = NULL,
  this_q2_test = NULL,
  return_this = "both"
)
```

Arguments

this_data_names
: Vector object with names of time series used in seasonal adjustment. This is a required entry.

this_acf_test
: List object with results from test of regARIMA residual ACF.

this_d11f_test
: List object with results from test of D11F.

this_spec_peak_test
: List object with results from testing for spectral peaks in the seasonally adjusted series.

this_spec_peak_ori_test
: List object with results from testing for spectral peaks in the original series.

this_qs_test
: List object with results from QS test.

this_qs_rsd_test
: List object with results from residual QS test.

this_qs_seasonal_test
: List object with results from seasonal QS test.

this_model_test
: List object with results from model diagnostics test.

this_sspan_test
: List object with results from sliding spans test.

this_m7_test
: List object with results from M7 test.

this_q2_test
: List object with results from Q2 test.

return_this
: Character string; what the function returns - `'why'` returns why the test failed or received a warning, `'test'` returns test results, or `'both'`. Default is `'both'`.

Details

Version 7.1, 2026-03-09

Value

A data frame with X-13 Diagnostics, with the elements not expressed as factors.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
unemp_seasonal_filter <- lapply(unemployment_list, function(x)
     try(optimal_seasonal_filter(x, use_msr = TRUE)))
unemp_seas_list <-
   seasonal::seas(unemployment_list,
      x11 = "", slidingspans = "",
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      forecast.maxlead=60,
      check.print = c( "pacf", "pacfplot" ),
      list = list(
         list(x11.seasonalma = unemp_seasonal_filter$n3000013),
         list(x11.seasonalma = unemp_seasonal_filter$n3000014),
         list(x11.seasonalma = unemp_seasonal_filter$n3000025),
         list(x11.seasonalma = unemp_seasonal_filter$n3000026)
      ))
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_acf <- lapply(unemp_seas_update, function(x)
  try(acf_test(x, return_this = 'both')))
unemp_d11f <- lapply(unemp_seas_update, function(x)
  try(d11f_test(x, p_level = 0.05, return_this = 'both')))
unemp_spec_peak <- lapply(unemp_seas_update, function(x)
  try(spec_peak_test(x, return_this = 'both')))
unemp_spec_peak_ori <- lapply(unemp_seas_update, function(x)
  try(spec_peak_test(x, this_spec = "spcori", return_this = 'both')))
unemp_qs <- lapply(unemp_seas_update, function(x)
  try(qs_test(x, test_full = FALSE, p_limit_fail = 0.01,
              p_limit_warn = 0.05, return_this = 'both')))
unemp_qs_rsd <- lapply(unemp_seas_update, function(x)
  try(qs_rsd_test(x, test_full = FALSE, p_limit_fail = 0.01,
                  p_limit_warn = 0.05, return_this = 'both')))
unemp_qs_seasonal <- lapply(unemp_seas_update, function(x)
  try(qs_seasonal_test(x, test_full = FALSE,
                       p_limit_pass = 0.01, p_limit_warn = 0.05,
                       robust_sa = FALSE, return_this = 'both')))
unemp_model <- lapply(unemp_seas_update, function(x)
  try(model_test(x, return_this = 'both')))
unemp_sspan <- lapply(unemp_seas_update, function(x)
  try(sspan_test(x, sf_limit = 15, change_limit = 35,
      return_this = 'both')))
unemp_m7 <- lapply(unemp_seas_update, function(x)
  try(mq_test(x, return_this = 'both')))
unemp_q2 <- lapply(unemp_seas_update, function(x)
  try(mq_test(x, this_label = 'Q2', return_this = 'both')))
unemp_names <- names(unemployment_list)
```

```
unemp_diag_df <-
    make_diag_df(unemp_names,
        this_acf_test = unemp_acf,
        this_d11f_test = unemp_d11f,
        this_spec_peak_test = unemp_spec_peak,
        this_spec_peak_ori_test = unemp_spec_peak_ori,
        this_qs_test = unemp_qs,
        this_qs_rsd_test = unemp_qs_rsd,
        this_qs_seasonal_test = unemp_qs_seasonal,
        this_model_test = unemp_model,
        this_sspan_test = unemp_sspan,
        this_m7_test = unemp_m7,
        this_q2_test = unemp_q2)
```

---

match_list *List element match*

Description

Returns element of list that matches this_string.

Usage

```
match_list(this_list, this_string = "fail", match_substring = FALSE)
```

Arguments

this_list	List of character strings.
this_string	Character string to match against elements of the list, ie, this_string = 'pass'. Default is 'fail'.
match_substring	Logical scalar; if TRUE, the function will consider a match to be a match of any substring in the string. The default is FALSE, where the string must match the entire string to be considered a match.

Details

Version 4.2, 2025-09-11

Value

A vector of list element names that match this_string. Users can specify that the whole string matches, or that a substring matches. If nothing matches, the function will output the string 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
unemp_seas <-
   seasonal::seas(unemployment_list, x11 = "",
      slidingspans = "",
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 60,
      check.print = c( "pacf", "pacfplot" ))
test_seas_update <-
   Filter(function(x) inherits(x, "seas"), unemp_seas)
test_acf_test <- lapply(test_seas_update, function(x)
     try(acf_test(x, return_this = 'test')))
test_acf_fail <- match_list(test_acf_test, 'fail')
test_acf_warn <- match_list(test_acf_test, 'warn')
test_acf_pass <- match_list(test_acf_test, 'pass')
```

`match_list_number` *Number of list element matches*

Description

Returns number of elements in list that matches `this_string`.

Usage

```
match_list_number(this_list, this_string = "fail", match_substring = FALSE)
```

Arguments

`this_list`	List of character strings.
`this_string`	Character string to match against elements of the list, ie, `this_string` = `'pass'`. Default is `'fail'`.
`match_substring`	Logical scalar; if `TRUE`, the function will consider a match to be a match of any substring in the string. The default is `FALSE`, where the string must match the entire string to be considered a match.

Details

Version 4.0, 2025-09-11

Value

The number of list items that match `this_string`. Users can specify that the whole string matches, or that a substring matches.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
unemp_seas <-
   seasonal::seas(unemployment_list, x11 = "",
      slidingspans = "",
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 60,
      check.print = c( "pacf", "pacfplot" ))
test_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas)
test_acf_test <-
     lapply(test_seas_update, function(x)
           try(acf_test(x, return_this = 'test')))
test_acf_number_fail <-
     match_list_number(test_acf_test, 'fail')
test_acf_number_warn <-
     match_list_number(test_acf_test, 'warn')
test_acf_number_pass <-
     match_list_number(test_acf_test, 'pass')
```

---

member_of_list *Member of list*

Description

Determines if a name is a member of a list.

Usage

```
member_of_list(this_list = NULL, this_name = NULL)
```

Arguments

this_list	A list of objects. This is a required entry.
this_name	character string; element name of this_list. This is a required entry.

Details

Version 1.5, 2026-04-16

Value

returns TRUE if this_name is an element of this_list, FALSE otherwise.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
emp_seas_list <-
   seasonal::seas(employment_list,
      slidingspans = "", x11 = "",
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead=36,
      forecast.save = "fct",
      check.print = c( "pacf", "pacfplot" ))
emp_seas_update <-
     Filter(function(x) inherits(x, "seas"), emp_seas_list)
if (member_of_list(emp_seas_update, 'n2000014')) {
    n2000014_fct <- seasonal::series(emp_seas_list$n2000014, "fct")
}
```

---

model_summary_list   *ARIMA model summary from a list*

Description

Generate a matrix that summarizes ARIMA models from a list of seas objects. Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
model_summary_list(
  seas_obj_list = NULL,
  add_diff = FALSE,
  return_data_frame = FALSE
)
```

Arguments

seas_obj_list	List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.
add_diff	logical scalar; add differencing information, if differences are included in model. Default is FALSE.
return_data_frame	logical scalar; if TRUE, return a data frame of the diagnostics, otherwise return a matrix. Default is FALSE.

Details

Version 3.0, 2026-02-26

Value

Matrix or data frame of ARMA model orders, nonseasonal and seasonal parameters for a given set of series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36, x11 = "",
                  check.print = c( "pacf", "pacfplot" ))
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_model_matrix <-
    model_summary_list(unemp_seas_update)
unemp_model_df <-
    model_summary_list(unemp_seas_update, add_diff = TRUE,
                       return_data_frame = TRUE)
```

---

model_test *Tests Time Series Model.*

Description

Tests whether the time series model has acceptable diagnostics.

Usage

```
model_test(
  seas_obj = NULL,
  t_value = 3,
  p_value = 0.05,
  otl_auto_limit = 5,
  otl_all_limit = 5,
  return_this = "test"
)
```

Arguments

seas_obj	Object generated by seas() of the seasonal package. (Sax and Eddelbuettel 2018). This is a required entry.
t_value	t-statistic limit for regressors. Default is 3.
p_value	p-value limit for regressors. Default is 0.01.
otl_auto_limit	Numeric object; limit for number of automatically identified outliers. Default is 4.

`otl_all_limit`	Numeric object; limit for number of outlier regressors. Default is `6`.
`return_this`	character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 5.0, 2026-02-26

Value

A text string denoting if the series passed or failed the tests of ARIMA diagnostics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead=20,
      check.print = c( "pacf", "pacfplot" ),
      x11="")
ukgas_model <-
   model_test(ukgas_seas,
      t_value=3.0,
      p_value=0.01,
      otl_auto_limit=4,
      otl_all_limit=6)
```

`model_test_list` *Tests Time Series Model.*

Description

Tests whether a list of seas objects have time series models that have acceptable diagnostics.

Usage

```
model_test_list(
  seas_obj_list = NULL,
  t_value = 3,
  p_value = 0.05,
  otl_auto_limit = 5,
  otl_all_limit = 5,
```

```
  return_this = "test",
  return_data_frame = FALSE
)
```

Arguments

seas_obj_list
: A list of seas objects for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.

t_value
: t-statistic limit for regressors. Default is `3`.

p_value
: p-value limit for regressors. Default is `0.01`.

otl_auto_limit
: Numeric object; limit for number of automatically identified outliers. Default is `4`.

otl_all_limit
: Numeric object; limit for number of outlier regressors. Default is `6`.

return_this
: character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'both'`.

return_data_frame
: logical scalar; if `TRUE`, return a data frame of the diagnostics, otherwise return a matrix. Default is `FALSE`.

Details

Version 1.2, 2026-04-17

Value

A list or data frame with text strings denoting if the each of the series passed or failed the tests of ARIMA diagnostics.

Author(s)

Brian C. Monsell, <`monsell.brian@bls.gov`> or <`bcmonsell@gmail.com`>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36, x11 = "",
                  check.print = c( "pacf", "pacfplot" ),
                  check.save = "acf",
                  series.save = "a1")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_model_df <-
   model_test_list(unemp_seas_list,
      t_value=3.0,
```

```
p_value=0.01,
otl_auto_limit=4,
otl_all_limit=6,
return_data_frame = TRUE)
```

---

model_test_why *Model Test Warning Message*

Description

Generates text on why a time series model is inadequate.

Usage

```
model_test_why(
  udg_list = NULL,
  t_value = 3,
  p_value = 0.05,
  otl_auto_limit = 5,
  otl_all_limit = 5,
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by udg() function of the seasonal package. (Sax and Eddelbuettel 2018). This is a required entry.
t_value	Numeric scalar; t-statistic limit for regressors. Default is 3.0.
p_value	Numeric scalar; p-value limit for regressors. Default is 0.01.
otl_auto_limit	Integer scalar; limit for number of automatically identified outliers. Default is 4.
otl_all_limit	Integer scalar; limit for number of automatically identified outliers. Default is 4.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is TRUE.

Details

Version 5.0, 2026-02-26

Value

A text string denoting why the series passed or failed a series of tests of ARIMA diagnostics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11="")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_model_why <-
   model_test_why(ukgas_udg,
      t_value = 3.0,
      p_value = 0.01,
      otl_auto_limit = 4,
      otl_all_limit = 6,
      return_both = TRUE)
```

---

mq_test — *Test X-11-ARIMA M and Q statistics*

Description

Generates a test for X-11-ARIMA M and Q statistics. For more information on the X-11-ARIMA quality control statistics, see Lothian and Morry (1978) and Ladiray and Quenneville (2001).

Usage

```
mq_test(
  seas_obj = NULL,
  this_label = "m7",
  this_fail_limit = 1.2,
  this_warn_limit = 0.8,
  return_this = "test"
)
```

Arguments

`seas_obj`	Object generated by `seas()` of the seasonal package. This is a required entry.
`this_label`	Character string; label for an M or Q statistic, such as `'M7'`, `'Q'`, or `'Q2'`. Default is `'M7'`.
`this_fail_limit`	Numeric scalar; value above which the M or Q statistic fails. Default is `1.2`.
`this_warn_limit`	Numeric scalar; value above which the M or Q statistic gives a warning message if it is less than `this_fail_limit`. Default is `0.8`
`return_this`	character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 4.0, 2026-02-26

Value

A text string denoting if series passes or has a warning for the quality control statistics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ladiray D, Quenneville B (2001). *Seasonal Adjustment with the X-11 Method*, volume 158. Springer, New York. Lecture Notes in Statistics.

Lothian J, Morry M (1978). “A Test of Quality Control Statistics for the X-11-ARIMA Seasonal Adjustment Program.” Research Paper, Seasonal Adjustment and Time Series Staff, Statistics Canada., https://www.census.gov/content/dam/Census/library/working-papers/1978/adrm/lothianmorry1978.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11="")
ukgas_q <-
   mq_test(ukgas_seas,
      this_label = 'q',
      return_this = 'both')
```

`norm_test` *Normality Tests for Time Series Models.*

Description

Tests different normality statistics available in X-13ARIMA-SEATS. Descriptions of these diagnostics can be found in Section 7.3 of the X-13ARIMA-SEATS Reference Manual U. S. Census Bureau (2025).

Usage

```
norm_test(seas_obj = NULL, this_norm = NULL, return_this = "test")
```

Arguments

`seas_obj`	Object generated by `seas()` of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_norm`	Type of normality statistic being extracted; permissable values are `'a'`, `'kurtosis'`, `'skewness'`. This is a required entry.
`return_this`	Character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 5.1, 2026-03-05

Value

A text string denoting whether the series passed or failed the specific normality test. If improper value is specified for this_norm, return NULL. If no statistic is found, return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11="")
ukgas_norm_a <- norm_test(ukgas_seas, 'a')
ukgas_norm_kurtosis <- norm_test(ukgas_seas, 'kurtosis')
ukgas_norm_skewness <- norm_test(ukgas_seas, 'skewness')
```

---

norm_test_why *Normality Test Warning Message*

Description

Generates message for why different normality statistics available in X-13ARIMA-SEATS fail. Descriptions of these diagnostics can be found in Section 7.3 of the X-13ARIMA-SEATS Reference Manual U. S. Census Bureau (2025).

Usage

```
norm_test_why(udg_list = NULL, this_norm = NULL, return_both = FALSE)
```

Arguments

- `udg_list` — List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
- `this_norm` — Character scalar inidicating type of normality statistic being extracted; permissable values are `'a'`, `'kurtosis'`, `'skewness'`. This is a required entry.
- `return_both` — Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 4.2, 2026-03-09

Value

A text string showing why a series failed the specific normality test.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(AirPassengers,
                  transform.function = "log",
                  forecast.maxlead=36,
                  check.print = c( "pacf", "pacfplot" ),
                  x11 = "",
                  slidingspans = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_norm_a_why <-
     norm_test_why(ukgas_udg, 'a', return_both = TRUE)
ukgas_norm_kurtosis_why <-
     norm_test_why(ukgas_udg, 'kurtosis', return_both = TRUE)
ukgas_norm_skewness_why <-
     norm_test_why(ukgas_udg, 'skewness', return_both = TRUE)
```

NP_test *Non-Parametric test from Maravall (2012)*

Description

Non-Parametric test for seasonality based on Kendal and Ord (1990), and originally due to Friedman from Maravall (2012). This code is adapted from `kendalls` subroutine in `ansub11.f` from the X-13ARIMA-SEATS source code (U. S. Census Bureau 2025).

Usage

```
NP_test(x = NULL)
```

Arguments

`x`	`ts` time series object. This is a required entry.

Details

Version 2.1, 2026-03-09

Value

List object with three elements: `ken` (test statistic), `df` (degrees of freedom), `cv` (test probability).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Kendal MG, Ord JK (1990). *Time Series*, 3rd edition. Charles Griffin, London.

Maravall A (2012). “Seasonality tests and automatic model identification in TRAMO-SEATS.” (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
NP_test_air <- NP_test(AirPassengers)
```

number_of_user_reg *Number of user defined regressors*

Description

Extreact the number of user-defined regressors in a seas objects series.

Usage

```
number_of_user_reg(seas_obj = NULL)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

Details

Version 2.0, 2026-02-26

Value

Number of user defined regressors in the model contained in seas_obj.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      check.save = "acf",
      x11="",
      slidingspans = "")
ukgas_num_user_reg <- number_of_user_reg(ukgas_seas)
```

---

optimal_seasonal_filter
Optimal X-11 seasonal moving average selection

Description

Determine the optimal X-11 seasonal moving average based on the value of the seasonal moving average coefficient from an airline model. For more information about X-13ARIMA-SEATS and seasonal package input, see U. S. Census Bureau (2025), and Sax and Eddelbuettel (2018).

Usage

```
optimal_seasonal_filter(
  this_series = NULL,
  aictest = NULL,
  model = "(0 1 1)(0 1 1)",
  variables = NULL,
  outlier = TRUE,
  trans = NULL,
  missing_code = NULL,
  this_xreg = NULL,
  dp_limits = TRUE,
  use_msr = FALSE,
  use_3x15 = TRUE
)
```

Arguments

this_series	A time series object. This is a required entry.
aictest	Character string with the entries for the regression.aictest argument to the seas function from the seasonal package. Default is NULL, AIC testing not done.
model	Character string with the entry for the arima.model argument to the seas function from the seasonal package. Default is "(0 1 1)(0 1 1)". Model should have a (0 1 1) seasonal term.
variables	Character string with the entries for the regression.variables argument to the seas function from the seasonal package. Default is NULL, no regressors added.
outlier	Logical scalar, if TRUE outlier identification is done in the call to the seas function from the seasonal package. Default is TRUE.
trans	Character scalar, a character string with the entry for the transform.function argument to the seas function. Default is NULL, and the entry auto will be used.
missing_code	Numeric scalar, a number with the entry for the series.missingcode argument to the seas function. Default is NULL, no missing value code is used.
this_xreg	Numeric matrix, a user defined regressor matrix to be used in the model estimation. Default is NULL, no user-defined regressors are used.
dp_limits	Logical scalar, if TRUE limits from Planas and Depoutot (2002) will be used to choose the moving average, else limits from Chu et al. (2012) will be used. Default is TRUE.

use_msr — Logical scalar, if TRUE result of MSR selection will be used if model cannot be estimated, otherwise function will return a NULL value. Default is FALSE.

use_3x15 — Logical scalar, if TRUE 3x15 seasonal filter will be returned if chosen, otherwise function will return a 3x9 value. Default is TRUE.

Details

Version 5.1, 2026-03-05

Value

The optimal X-11 seasonal filter, unless the airline model cannot be estimated.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Chu Y, Tiao GC, Bell WR (2012). "A Mean Squared Error Criterion for Comparing X-12-ARIMA and Model-Based Seasonal Adjustment Filters." *Taiwan Economic Forecast and Policy*, **43**(1), 1–32. https://www.census.gov/content/dam/Census/library/working-papers/2012/adrm/chutiaobelltefp.pdf.

Planas C, Depoutot R (2002). "Controlling Revisions in ARIMA-Model-Based Seasonal Adjustment." *Journal of Time Series Analysis*, **23**, 193–214. doi:10.1111/14679892.00262.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
this_seasonal  <-
   optimal_seasonal_filter(shoes2008,
      aictest = c("td", "easter"),
      use_msr = TRUE)
this_seasonal2 <-
   optimal_seasonal_filter(shoes2008,
      aictest = c("td", "easter"),
      dp_limits = FALSE,
      use_msr = TRUE)
```

`overall_seasonal_test_1`
First overall sasonality test from Maravall (2012).

Description

Conduct the first overall test for seasonality as laid out in Maravall (2012).

Usage

```
overall_seasonal_test_1(
  seas_obj = NULL,
  this_series = "a1",
  take_log = TRUE,
  this_ori_qs = "qsori",
  return_data_frame = FALSE
)
```

Arguments

`seas_obj`	A `seas` object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`this_series`	Character string; the table used to generate the AR(30) spectrum. Default is `"a1"`.
`take_log`	Logical scalar; indicates if the AR spectrum is generated from the log of the data. Default is `TRUE`.
`this_ori_qs`	Character string; code for which original series QS will be used in this analysis. Default is `"qsori"` (original series, full span); other choices are `"qssorievadj"` (original series adjusted for extreme values, short span), `"qsorievadj"` (original series adjusted for extreme values, full span), and `"qssori"` (original series, short span). For more information on the QS statistic, see Bell et al. (2022).
`return_data_frame`	Logical scalar; if `TRUE`, return a data frame of the diagnostics, otherwise return a list. Default is `FALSE`.

Details

Version 4.6, 2026-04-16

Value

A list or data frame with 3 elements: `QStest` (test probability for QS), `NPtest` (test probability for NP), and `result` (character string with test result - possible values of either `"evidence of seasonality"` and `"no evidence of seasonality"`).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      arima.model="(0 1 1)(0 1 1)",
      forecast.maxlead = 36,
      slidingspans = "",
      transform.function = "log",
      series.save = "a1")
first_test <-
     overall_seasonal_test_1(air_seas, return_data_frame = TRUE)
```

---

overall_seasonal_test_1_list

First overall sasonality test from Maravall (2012) applied to a list.

Description

Conduct the first overall test for seasonality as laid out in Maravall (2012) applied to a list of seasonal objects. Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
overall_seasonal_test_1_list(
  seas_obj_list = NULL,
  this_series = "a1",
  take_log = TRUE,
  this_ori_qs = "qsori",
  return_data_frame = FALSE
)
```

Arguments

seas_obj_list	A list of seas objects for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; the table used to generate the AR(30) spectrum. Default is "a1".
take_log	Logical scalar; indicates if the AR spectrum is generated from the log of the data. Default is TRUE.

this_ori_qs
: Character string; code for which original series QS will be used in this analysis. Default is "qsori" (original series, full span); other choices are "qssorievadj" (original series adjusted for extreme values, short span), "qsorievadj" (original series adjusted for extreme values, full span), and "qssori" (original series, short span). For more information on the QS statistic, see Bell et al. (2022).

return_data_frame
: Logical scalar; if TRUE, return a data frame of the diagnostics, otherwise return a matrix. Default is FALSE.

Details

Version 1.4, 2026-04-16

Value

A list or data frame of seasonality diagnostics for a set of series. Each object will contain 3 elements: QStest (test probability for QS), NPtest (test probability for NP), and result (character string with test result - possible values of either "evidence of seasonality" and "no evidence of seasonality").

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36, x11 = "",
                  check.print = c( "pacf", "pacfplot" ),
                  check.save = "acf",
                  series.save = "a1")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
first_test_df <-
     overall_seasonal_test_1_list(unemp_seas_update,
                                  return_data_frame = TRUE)
```

overall_seasonal_test_2
Second overall sasonality test from Maravall (2012).

Description

Conduct the second overall test for seasonality as laid out in Maravall (2012).

Usage

```
overall_seasonal_test_2(
  seas_obj = NULL,
  this_ar_spec_cv = NULL,
  n_sim = 1e+05,
  this_series = "b1",
  this_ori_qs = "qssorievadj",
  take_log = TRUE,
  take_diff = TRUE,
  use_seasonal = TRUE,
  return_data_frame = FALSE
)
```

Arguments

- `seas_obj` A `seas` object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
- `this_ar_spec_cv` List object with two elements - 99 and 95 percent critical values for the frequencies of the AR(30) spectrum as generated by the `gen_ar_spec_cv` function.
- `n_sim` Integer scalar; number of simulations; this option is used only if `this_ar_spec_cv` is not specified. Default is `100000`.
- `this_series` Character string; the table used to generate the AR(30) spectrum. Default is `"b1"`.
- `this_ori_qs` Character string; code for which original series QS will be used in this analysis. Default is `"qssorievadj"` (original series adjusted for extreme values, short span); other choices are `"qsori"` (original series, full span), `"qsorievadj"` (original series adjusted for extreme values, full span), and `"qssori"` (original series, short span). For more information on the QS statistic, see Bell et al. (2022).
- `take_log` Logical scalar; indicates if the AR spectrum is generated from the log of the data. Default is `TRUE`.
- `take_diff` Logical scalar; indicates if the data is differenced before the AR spectrum is generated. Default is `TRUE`.
- `use_seasonal` Logical scalar; if TRUE, the `seasonal` regressor is used; otherwise, use the sine-cosine trignonmetric regressors generated by `sincos`.
- `return_data_frame` Logical scalar; if `TRUE`, return a data frame of the diagnostics, otherwise return a matrix. Default is `FALSE`.

Details

Version 6.6, 2026-04-16

Value

A list or data frame with 5 elements: `QStest` (test probability for QS), `NPtest` (test probability for NP), `Ftest` (test probability for model based seasonal F-test), `Spectrum` (character string with test result - possible values of either `"evidence of seasonal peak"`, `"no evidence of seasonal peak"`), and `Result` (character string with test result - possible values of either `"strong seasonal"`, `"weak seasonal"`, `"no seasonal"`.

Author(s)

Brian C. Monsell, `<monsell.brian@bls.gov>` or `<bcmonsell@gmail.com>`

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      series.save = "b1",
      transform.function = "log",
      arima.model="(0 1 1)(0 1 1)",
      forecast.maxlead = 36,
      slidingspans = "")
ar30_spec_cv <- gen_ar_spec_cv(1000, 97, 12)
second_test <-
    overall_seasonal_test_2(air_seas, ar30_spec_cv,
                            return_data_frame = TRUE)
```

---

overall_seasonal_test_2_list

Second overall sasonality test from Maravall (2012) applied to a list.

Description

Conduct the second overall test for seasonality as laid out in Maravall (2012) to a list of seasonal objects. Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
overall_seasonal_test_2_list(
  seas_obj_list = NULL,
  this_ar_spec_cv = NULL,
  n_sim = 1e+05,
  this_series = "b1",
  this_ori_qs = "qssorievadj",
  take_log = TRUE,
  take_diff = TRUE,
  use_seasonal = TRUE,
  return_data_frame = FALSE
)
```

Arguments

- `seas_obj_list` A list of `seas` objects generated by the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
- `this_ar_spec_cv` List object with two elements - 99 and 95 percent critical values for the frequencies of the AR(30) spectrum as generated by the `gen_ar_spec_cv` function.
- `n_sim` Integer scalar; number of simulations; this option is used only if `this_ar_spec_cv` is not specified. Default is `100000`.
- `this_series` Character string; the table used to generate the AR(30) spectrum. Default is `"b1"`.
- `this_ori_qs` Character string; code for which original series QS will be used in this analysis. Default is `"qssorievadj"` (original series adjusted for extreme values, short span); other choices are `"qsori"` (original series, full span), `"qsorievadj"` (original series adjusted for extreme values, full span), and `"qssori"` (original series, short span). For more information on the QS statistic, see Bell et al. (2022).
- `take_log` Logical scalar; indicates if the AR spectrum is generated from the log of the data. Default is `TRUE`.
- `take_diff` Logical scalar; indicates if the data is differenced before the AR spectrum is generated. Default is `TRUE`.
- `use_seasonal` Logical scalar; if TRUE, the `seasonal` regressor is used; otherwise, use the sine-cosine trignonmetric regressors generated by `sincos`.
- `return_data_frame` Logical scalar; if `TRUE`, return a data frame of the diagnostics, otherwise return a matrix. Default is `FALSE`.

Details

Version 1.4, 2026-04-15

Value

A list or data frame of seasonality diagnostics for a set of series. Each object will contain 5 elements: `QStest` (test probability for QS), `NPtest` (test probability for NP), `Ftest` (test probability for model based seasonal F-test), `Spectrum` (character string with test result - possible values of either `"evidence of seasonal peak"`, `"no evidence of seasonal peak"`), and `Result` (character string with test result - possible values of either `"strong seasonal"`, `"weak seasonal"`, `"no seasonal"`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list, slidingspans = "",
                  transform.function = "log",
                  outlier.types = "all",
                  arima.model = "(0 1 1)(0 1 1)",
                  forecast.maxlead=36, x11 = "",
                  check.print = c( "pacf", "pacfplot" ),
                  check.save = "acf",
                  series.save = "a1")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
second_test <-
    overall_seasonal_test_2_list(unemp_seas_update,
                                 return_data_frame = TRUE)
```

---

process_list   *Process list object of numbers*

Description

Process list object of numbers and return names of elements that are either greater than or less than a limit.

Usage

```
process_list(
  this_list = NULL,
  this_limit = NULL,
  abs_value = FALSE,
  greater_than = TRUE
)
```

Arguments

this_list	List of numeric values. The elements should be scalars, not arrays or text. This is a required entry.
this_limit	Numeric scalar which serves as the limit of the numbers stored in this_list. This is a required entry.
abs_value	Logical scalar that indicates whether the absolute value is taken of the numbers before the comparison is made. Default is FALSE.
greater_than	Logical object that specified whether the element names returned are greater than or less than the limit specified in this_limit. Default is TRUE.

Details

Version 2.3, 2026-02-26

Value

A vector of list element names where the value in this_list is greater than or less than the limit specified in this_limit. If nothing matches, the function will output the string 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
emp_seas_list <-
  seasonal::seas(employment_data_mts,
                 slidingspans = "", forecast.maxlead=36,
                 arima.model = "(0 1 1)(0 1 1)",
                 transform.function = "log", x11 = "",
                 check.print = c( "pacf", "pacfplot" ))
emp_seas_update <-
    Filter(function(x) inherits(x, "seas"), emp_seas_list)
m7_key          <- get_mq_key('M7')
emp_m7_list     <-
   lapply(emp_seas_update, function(x)
          try(get_udg_entry(x, m7_key)))
emp_m7_pass     <-
   process_list(emp_m7_list,
      this_limit = 1.0,
      abs_value = TRUE,
      greater_than = FALSE)
```

---

proc_outlier — *Extract dates from outlier text*

Description

Process name of outlier regressor to extract the dates associated with the outlier.

Usage

```
proc_outlier(this_outlier = NULL, this_freq = 12, add_type = TRUE)
```

Arguments

`this_outlier`	Character string; outlier regressor. This is a required entry.
`this_freq`	integer scalar; time series frequency. Default is 12.
`add_type`	logical scalar; determines if type of outlier is added to the output. Default is `TRUE`.

Details

Version 2.2, 2026-02-26

Value

List of either year and month/quarter of outlier, or year and month/quarter of start and end of outlier.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      regression.aictest = 'td',
      forecast.maxlead=36,
      check.print = c( "pacf", "pacfplot" ),
      x11="",
      slidingspans = "",)
this_auto_outlier <- get_auto_outlier_string(air_seas)
this_outlier      <- proc_outlier(this_auto_outlier)
```

---

qs_fail_why *QS diagnostic failure message*

Description

Generates text explaining why the QS diagnostic failed or generated a warning. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_fail_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_fail = 0.01,
  robust_sa = TRUE,
  return_both = FALSE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`test_full`	Logical scalar indicating whether to test the full series span. Default is `TRUE`.
`test_span`	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is `TRUE`.
`p_limit_fail`	Numeric scalar; P-value limit for QS statistic for failure. Default is `0.01`.
`robust_sa`	Logical scalar indicating if SA or irregular series adjusted for extremes is included in testing. Default is `TRUE`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 5.1, 2026-03-05

Value

A text string denoting why the series failed the tests of QS diagnostics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_test <-
```

```
qs_fail_why(ukgas_udg,
   test_full = FALSE,
   p_limit_fail = 0.01,
   return_both = TRUE)
```

---

qs_rsd_fail_why *QS diagnostic for regarima residuals failure message*

Description

Generates text explaining why the QS diagnostic failed or generated a warning for regARIMA residuals. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_rsd_fail_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_fail = 0.01,
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by udg() function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span. Default is TRUE.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
p_limit_fail	Numeric scalar; P-value limit for QS statistic for warning. Default is 0.01.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is FALSE.

Details

Version 5.1, 2026-03-05

Value

A text string denoting why the series failed the QS test of regARIMA residuals.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). “Identifying Seasonality.” BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). “Seasonality tests and automatic model identification in TRAMO-SEATS.” (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_rsd_fail_why <-
  qs_rsd_fail_why(ukgas_udg,
      test_full = FALSE,
      p_limit_fail = 0.01,
      return_both = TRUE)
```

---

qs_rsd_test *QS diagnostic test*

Description

Tests using the QS diagnostic developed by Maravall. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_rsd_test(
  seas_obj = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_fail = 0.01,
  p_limit_warn = 0.05,
  return_this = "test"
)
```

Arguments

seas_obj	seas object generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span. Default is TRUE.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
p_limit_fail	Numeric scalar; P-value limit for QS statistic for failure. Default is 0.01.
p_limit_warn	Numeric scalar; P-value limit for QS statistic for warning. Default is 0.05.
return_this	character string; what the function returns - 'test' returns test results, 'why' returns why the test failed or received a warning, or 'both'. Default is 'test'.

Details

Version 5.1, 2026-03-05

Value

A text string denoting if the regarima residuals passed or failed tests for residual seasonality using the QS diagnostics. Can test the entire series or the last 8 years or both.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_qs_test_rsd <-
   qs_rsd_test(ukgas_seas,
```

```
        test_full = FALSE,
        p_limit_fail = 0.01,
        p_limit_warn = 0.05,
        return_this = 'both')
```

---

qs_rsd_warn_why *Residual QS diagnostic warning message.*

Description

Generates text explaining why the QS diagnostic failed or generated a warning for regARIMA residuals. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_rsd_warn_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_warn = 0.05,
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span. Default is `TRUE`.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is `TRUE`.
p_limit_warn	Numeric scalar; P-value limit for QS statistic for warning. Default is `0.05`.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 4.1, 2026-03-05

Value

A text string denoting why the series generated a warning message for the QS of regARIMA residuals.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_rsd_warn <-
   qs_rsd_warn_why(ukgas_udg,
      test_full = FALSE,
      p_limit_warn = 0.05,
      return_both = TRUE)
```

---

qs_seasonal_fail_why *QS Test for original series*

Description

Tests using the QS diagnostic developed by Maravall to determine if the original series is seasonal. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_seasonal_fail_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_warn = 0.05,
  robust_sa = TRUE,
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by udg() function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span Default is TRUE.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
p_limit_warn	Numeric scalar; P-value limit for QS statistic for warning Default is 0.05.
robust_sa	Logical scalar indicating if original series adjusted for extremes is included in testing. Default is FALSE.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is FALSE.

Details

Version 5.1, 2026-03-05

Value

A text string denoting why the series passed or failed the QS diagnostic.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_fail_seasonal <-
```

```
qs_seasonal_fail_why(ukgas_udg,
   test_full = FALSE,
   p_limit_warn = 0.05,
   robust_sa = FALSE,
   return_both = FALSE)
```

---

qs_seasonal_test *QS seasonal tests*

Description

Tests using the QS diagnostic developed by Maravall to determine if the original series is seasonal. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_seasonal_test(
  seas_obj = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_pass = 0.01,
  p_limit_warn = 0.05,
  robust_sa = TRUE,
  return_this = "test"
)
```

Arguments

seas_obj	seas object generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span Default is TRUE.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
p_limit_pass	Numeric scalar; P-value limit for QS statistic for passing Default is 0.01.
p_limit_warn	Numeric scalar; P-value limit for QS statistic for warning Default is 0.05.
robust_sa	Logical scalar indicating if original series adjusted for extremes is included in testing. Default is TRUE.
return_this	Character string; what the function returns - 'test' returns test results, 'why' returns why the test failed or received a warning, or 'both'. Default is 'test'.

Details

Version 5.2, 2026-03-05

Value

A text string denoting if the series passed or failed tests for seasonality using the QS diagnostics. Can test the entire series or the last 8 years or both.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",,
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead = 20,
      x11 = "")
ukgas_qs_test_seasonal <-
   qs_seasonal_test(ukgas_seas,
      test_full = FALSE,
      p_limit_pass = 0.01,
      p_limit_warn = 0.05,
      robust_sa=FALSE,
      return_this = 'both')
```

---

qs_seasonal_warn_why   *Warning or error messages for QS seasonal diagnostic*

Description

Tests using the QS diagnostic developed by Maravall to determine if the original series is seasonal. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_seasonal_warn_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
```

```
  p_limit_pass = 0.05,
  robust_sa = TRUE,
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by udg() function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
test_full	Logical scalar indicating whether to test the full series span Default is TRUE.
test_span	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
p_limit_pass	Numeric scalar; P-value limit for QS statistic for passing. Default is 0.05.
robust_sa	Logical scalar indicating if original series adjusted for extremes is included in testing. Default is FALSE.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is FALSE.

Details

Version 5.2, 2026-03-05

Value

A text string denoting why the series had a warning message from the tests for seasonality using the QS diagnostics. Can test the entire series or the last 8 years or both.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
```

```
        arima.model = "(0 1 1)(0 1 1)",,
        check.print = c( "pacf", "pacfplot" ),
        forecast.maxlead = 20,
        x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_warn_seasonal <-
   qs_seasonal_warn_why(ukgas_udg,
        test_full = FALSE,
        p_limit_pass = 0.025,
        robust_sa=FALSE,
        return_both = TRUE)
```

---

qs_series *QS diagnostic test on a number of series.*

Description

Apply QS Tests to a list of seas objects. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022). Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
qs_series(
  seas_obj_list = NULL,
  this_stat = "qsori",
  less_than = TRUE,
  p_limit = 0.01
)
```

Arguments

seas_obj_list	List object of seas objects generated from a call of seas on a list consisting of individual time series.
this_stat	Character string that specifies which QS statistic is being tested. Allowable values are 'qsori', 'qsorievadj', 'qsrsd', 'qssadj', 'qssadjevadj', 'qsirr', 'qsirrevadj', 'qssori', 'qssorievadj', 'qssrsd', 'qsssadj', 'qsssadjevadj', 'qssirr', 'qssirrevadj'. Default is "qsori".
less_than	Logical scalar which indicates if the test is going to be QS less than p_limit (less_than = TRUE) or QS greater than p_limit (less_than = FALSE). Default is TRUE.
p_limit	Numeric scalar; P-value limit for QS statistic. Default is 0.01.

Details

Version 4.1, 2026-03-05

Value

A vector of list element names that have the given QS statistic either less than or greater than the given P-value limit. If nothing matches, the function will output the string 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
emp_seas_list <-
   seasonal::seas(employment_list,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      regression.aictest = NULL,
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead=60,
      x11 = "",
      slidingspans = "")
test_seas_update <-
   Filter(function(x) inherits(x, "seas"), emp_seas_list)
test_qs_names <-
   qs_series(test_seas_update,
      this_stat = 'qssori',
      less_than = FALSE,
      p_limit = 0.025)
```

`qs_test` *QS Test for residual seasonality*

Description

Tests using the QS diagnostic developed by Maravall on seasonally adjusted series and the irregular component. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_test(
  seas_obj = NULL,
  test_full = TRUE,
```

```
  test_span = TRUE,
  p_limit_fail = 0.01,
  p_limit_warn = 0.05,
  robust_sa = TRUE,
  return_this = "test"
)
```

Arguments

`seas_obj`	seas object generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
`test_full`	Logical scalar indicating whether to test the full series span. Default is TRUE.
`test_span`	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is TRUE.
`p_limit_fail`	Numeric scalar; P-value limit for QS statistic for failure. Default is 0.01.
`p_limit_warn`	Numeric scalar; P-value limit for QS statistic for warning. Default is 0.05.
`robust_sa`	Logical scalar indicating if SA or irregular series adjusted for extremes is included in testing. Default is TRUE.
`return_this`	character string; what the function returns - 'test' returns test results, 'why' returns why the test failed or received a warning, or 'both'. Default is 'test'.

Details

Version 5.2, 2026-03-05

Value

A text string denoting if the series passed or failed tests for residual seasonality using the QS diagnostics. Can test the entire series or the last 8 years or both.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",,
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead = 20,
      x11 = "")
ukgas_qs_test <-
   qs_test(ukgas_seas,
      test_full = FALSE,
      p_limit_fail = 0.01,
      p_limit_warn = 0.05,
      return_this = 'both')
```

`qs_warn_why` *Warning message for QS Test for residual seasonality*

Description

Generates text explaining why the QS diagnostic generated a warning. For details on the QS test, see Section 7.17 of U. S. Census Bureau (2025), Maravall (2012), and Bell et al. (2022).

Usage

```
qs_warn_why(
  udg_list = NULL,
  test_full = TRUE,
  test_span = TRUE,
  p_limit_warn = 0.05,
  robust_sa = TRUE,
  return_both = FALSE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`test_full`	Logical scalar indicating whether to test the full series span Default is `TRUE`.
`test_span`	Logical scalar indicating whether to test the final 8-year span used by the spectrum diagnostic. Default is `TRUE`.
`p_limit_warn`	Numeric scalar; P-value limit for QS statistic for warning. Default is `0.05`.
`robust_sa`	Logical scalar indicating if SA or irregular series adjusted for extremes is included in testing. Default is `TRUE`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 5.2, 2026-03-05

Value

A text string denoting why the series had a warning message from the tests for seasonality using the QS diagnostics. Can test the entire series or the last 8 years or both.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Maravall A (2012). "Seasonality tests and automatic model identification in TRAMO-SEATS." (Version: November 12, 2012) Bank of Spain, Servicio de Estudios.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",,
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead = 20,
      x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_qs_test <-
   qs_warn_why(ukgas_udg,
      test_full = FALSE,
      p_limit_warn = 0.025,
      return_both = TRUE)
```

---

replace_na *Replace NA*

Description

Replace NA in a vector with a string or number.

Usage

```
replace_na(this_vec = NULL, replace_string = "NA", replace_number = NULL)
```

Arguments

`this_vec`	Vector object. This is a required entry.
`replace_string`	Character scalar which replaces the NAs in the vector. Default is `'NA'`.
`replace_number`	Number which replaces the `NA`s in the vector. Default is the `NA` is replaced by the string in `replace_string`.

Details

Version 2.5, 2026-03-03

Value

A vector with all `NA`s replaced by either a character string or a number.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
sample_vec <- c(rnorm(25), NA, rnorm(24))
sample_vec_missing  <-
   replace_na(sample_vec, replace_string = 'Missing')
sample_vec_missing_number  <-
   replace_na(sample_vec, replace_number = -9999)
```

`r_terror` *TERROR for R*

Description

A function that duplicates the functionality of the TERROR software (Caporello and Maravall 2003; Maravall 2007) that performs quality control on time series based on one step ahead forecasts. This function calls the `seasonal` package (Sax and Eddelbuettel 2018). Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
r_terror(
  this_series = NULL,
  max_lead = 36,
  log_transform = TRUE,
  aictest = NULL,
  terror_lags = 1
)
```

Arguments

`this_series`	Time series array. This is a required entry.
`max_lead`	Number of forecasts generated by the seas run. Default is `36`.
`log_transform`	Logical scalar, if TRUE `transform.function` will be `auto` will be used. Default is `TRUE`.
`aictest`	Character string with the entries for the `regression.aictest` argument to the `seas` function from the `seasonal` package. Default is `NULL`.
`terror_lags`	Integer scalar for number of forecast lags from the end of series we'll collect t-statistics. Must be either 1, 2, or 3. Default is `1`.

Details

Version 4.0, 2026-03-03

Value

T-statistics generated by out of sample forecast error for the last 1 to 3 observation of each series in the list.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Caporello G, Maravall A (2003). "A Tool for Quality Control of Time Series Data. Program TERROR." Occasional Paper 0301, Servicio de Estudios, Bank of Spain, https://www.bde.es/f/webbde/SES/Secciones/Publicaciones/PublicacionesSeriadas/DocumentosOcasionales/03/Fic/do0301e.pdf.

Maravall A (2007). "Large Scale Applied Time Series Analysis with Program TSW (TRAMO-SEATS for Windows)." Proceedings for the Federal Council for Statistical Methodology, https://nces.ed.gov/FCSM/pdf/2007FCSM_Maravall-TD.pdf.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_terror <-
   r_terror(AirPassengers,
      log_transform = TRUE,
      aictest = c('td', 'easter'),
      terror_lags = 3)
```

`r_terror_list` *TERROR for R (applied to a list of series)*

Description

Function that duplicates the functionality of the TERROR software (Caporello and Maravall 2003; Maravall 2007) that performs quality control on time series based on one step ahead forecasts. The data is stored in a list of time series (`ts`) objects. This function calls the `seasonal` package (Sax and Eddelbuettel 2018).

Usage

```
r_terror_list(
  this_data_list = NULL,
  this_lead = 36,
  this_log = TRUE,
  this_aictest = NULL,
  this_terror_lags = 1
)
```

Arguments

`this_data_list` List of time series (all series in list should be the same frequency and have the same ending date.)

`this_lead` Number of forecasts generated by the `seas` run. Default is `36`.

`this_log` logical scalar, if TRUE `transform.function` will be set to `log` in the call to the `seas` function from the `seasonal` package, otherwise `auto` will be used. Default is `TRUE`.

`this_aictest` a character string with the entries for the `regression.aictest` argument to the `seas` function from the `seasonal` package. Default is `NULL`.

`this_terror_lags`
Integer scalar for number of forecast lags from the end of series where t-statistics are collected. Must be either 1, 2, or 3. Default is `1`.

Details

Version 4.0, 2026-03-03

Value

list of t-statistics generated by out of sample forecast error for the last 1 to 3 observation of each series in the list.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Caporello G, Maravall A (2003). "A Tool for Quality Control of Time Series Data. Program TERROR." Occasional Paper 0301, Servicio de Estudios, Bank of Spain, https://www.bde.es/f/webbde/SES/Secciones/Publicaciones/PublicacionesSeriadas/DocumentosOcasionales/03/Fic/do0301e.pdf.

Maravall A (2007). "Large Scale Applied Time Series Analysis with Program TSW (TRAMO-SEATS for Windows)." Proceedings for the Federal Council for Statistical Methodology, https://nces.ed.gov/FCSM/pdf/2007FCSM_Maravall-TD.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
emp_terror <-
   r_terror_list(employment_list,
      this_log = FALSE,
      this_lead = 60,
      this_terror_lags = 3)
```

`save_metafile` *Generate X-13ARIMA-SEATS metafile*

Description

Generates external input metafile for spec files generated from a list of seas objects. For information on input metafiles used with X-13ARIMA-SEATS, see Section 2.5 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025). Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
save_metafile(
  seas_obj_list = NULL,
  this_name_vec = NULL,
  metafile_name = NULL,
  this_directory = NULL,
  this_spec_directory = NULL,
  this_output_code = NULL,
  this_output_directory = NULL,
  include_directory = FALSE
)
```

Arguments

seas_obj_list	List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.
this_name_vec	Vector of character string; vector of series names from the list of seas objects that will be saved. Default is all elements of the seasonal object list seas_obj_list are saved.

metafile_name Character string; base name of metafile to be generated. If not specified, use name of list input as metafile name. Note - do not specify the ".mta" file extension.

this_directory Optional directory where the meta file is stored. If not specified, the metafile will be saved in the current working directory.

this_spec_directory
Optional directory where the spec files are stored. If not specified, the spec files are saved in the current working directory.

this_output_code
Optional character code added to the end of the names in this_name_vec to form the output name. If not specified, the metafile will not have alternate output file name(s).

this_output_directory
Optional directory where the output files are stored. If not specified, the output files are saved in the current working directory.

include_directory
Logical scalar; if TRUE, include directory specified in this_directory with file name output. Otherwise, output only names in this_name_vec. Default is FALSE. Note that the argument this_directory must also be specified.

Details

Version 4.2, 2026-04-14

Value

Generates metafile that can be used directly with the X-13ARIMA-SEATS program.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C (2024). “Adjusting Multiple Series — cran.r-project.org.” [Accessed 20-02-2026], https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead=36,
      x11 = "",
      slidingspans = "")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
```

```
## Not run: save_metafile(unemp_seas_update, metafile_name = 'unemp',
              this_directory = 'X:\\seasonalAdj\\testing\\sautilities',
              this_spec_directory = 'X:\\seasonalAdj\\testing\\spc',
              this_output_directory = 'X:\\seasonalAdj\\testing\\out',
              include_directory = TRUE)
## End(Not run)
```

`save_seas_object` *Save seas objects*

Description

Saves the seas command to reproduce the seas object seas_obj into the file file_name.r. More information on the seasonal package can be found at Sax and Eddelbuettel (2018).

Usage

```
save_seas_object(
  seas_obj = NULL,
  file_name = NULL,
  series_name = NULL,
  data_list = NULL,
  list_element = NULL,
  user_reg = NULL,
  this_window = FALSE,
  this_directory = NULL,
  this_sep = "_",
  print_out = FALSE
)
```

Arguments

seas_obj	seas object. This is a required entry.
file_name	Character string; file name where seas object is stored. Default is the name of the seasonal object.
series_name	Character string; name of time series object used by the seas object. Default is the name of the seasonal object.
data_list	Character string; name of the list object that holds data.
list_element	Character string; name of the list element used as data. Default is the name of the seasonal object.
user_reg	Character string; name of a time series matrix containing user defined regressors. There is no default. If not set, will set variables related to user defined regressors to NULL in the static version of the seas object.
this_window	Logical indicator variable; determines if a span of the window() function. If FALSE, the entire series will be used in the saved file.
this_directory	Character string; optional directory where the spec file is stored.
this_sep	Character string; separator between elements of the file name. Default is "_".
print_out	Logical indicator variable; determines if a call to the out() function of the seasonal package is put at the end of the script. If FALSE, the out() function is commented out. Default is FALSE.

Details

Version 10.1, 2026-03-05

Value

Stores the seas command to reproduce the seas object seas_obj into the file file_name.r - if file_name is not specified, the name of the seasonal object will be used to form the output file name.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
## Not run: save_seas_object(ukgas_seas,
            file_name = "ukgas_seas", series_name = "ukgas",
            print_out = TRUE)
## End(Not run)
```

---

save_series *Save Series*

Description

Save an R object with time series attributes (ts) to an external ASCII file in X-13ARIMA-SEATS' datevalue format or X-13ARIMA-SEATS' datevalue format. For more information on the datevalue format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
save_series(this_series = NULL, this_file = NULL, save_dv = TRUE)
```

Arguments

this_series	Time series array to be saved. This is a required entry.
this_file	Character string; name of file time series array to be saved to. This is a required entry.
save_dv	Logical scalar; if TRUE, time series object is stored in datevalue format; else, the time series object is stored in x13save format.

Details

Version 3.0, 2026-07-01

Value

File with time series data in the specified format will be produced.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Monsell BC, Lytras D, Findley D (2016). "Getting Started with X-13ARIMA-SEATS Input Files." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/gettingstartedx13-winx13.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
ukgas_sa <- seasonal::final(ukgas_seas)
## Not run: save_series(ukgas_sa, 'ukgas_sa.txt')
```

---

save_spec_file *Save spec file representation of* seas *object*

Description

Stores the spec file representation of the seas object seas_obj into the file file_name.spc.

Usage

```
save_spec_file(
  seas_obj = NULL,
  file_name = NULL,
  this_directory = NULL,
  this_data_directory = NULL,
  data_file_name = NULL,
  xreg_file_name = NULL,
  this_user_name = NULL,
  this_title = NULL
)
```

Arguments

seas_obj
: A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

file_name
: Character string; file name where seas object is stored. Default is the name of the seasonal object.

this_directory
: Character string; optional directory where the spec file is stored.

this_data_directory
: Character string; optional directory where the data files are stored. Default is no change in file entry in the spec file.

data_file_name
: Character string; optional external file name where the data file is stored. Path should be included with file name if data file is not in working directory; quotes will be added by the routine. Default is no change in file entry in the spec file.

xreg_file_name
: Character string; optional external file name where user defined regressors are stored. Path should be included with file name if the data file is not in the working directory; quotes will be added by the routine. Default is no change in file entry in the spec file.

this_user_name
: Vector of character strings; optional names for the user-defined regressors. Should only appear if xreg_file_name is specified.

this_title
: Character string; optional custom title; quotes will be added by the routine. Default is no change in title entry in the spec file.

Details

Version 5.1, 2026-03-05

Value

Stores the spec file representation of the seas object seas_obj into the file file_name.spc.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 20,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "")
## Not run: save_spec_file(ukgas_seas, 'ukgas',
                           data_file_name = "ukgas.dat",
                           this_directory = "X:\\seasonalAdj\\testing\\sautilities",
                           this_title = "Production run for quarterly UK Gas")
## End(Not run)
```

save_spec_file_vec *Save List of Seas Objects*

Description

Stores each element of a list of seas object this_seas_object into separate files. Information on generating adjustments of multiple time series stored in a list object with the seasonal package can be found in Sax (2024).

Usage

```
save_spec_file_vec(
  seas_obj_list = NULL,
  this_name_vec = NULL,
  this_directory = NULL,
  this_data_directory = NULL,
  this_ext = ".dat",
  this_title_list = NULL,
  this_title_base = NULL,
  this_xreg_list = NULL,
  this_user_list = NULL,
  make_metafile = FALSE,
  this_metafile_name = NULL,
  this_meta_directory = NULL,
  this_output_directory = NULL,
  include_directory = FALSE
)
```

Arguments

seas_obj_list	List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.
this_name_vec	Vector of character string; vector of series names from the list of seas objects that will be saved. Default is all elements of the seasonal object list seas_obj_list are saved.
this_directory	Character string; optional directory where the spec file is stored.
this_data_directory	Character string; optional directory where the data files are stored. Data files are assumed to have the same names as in this_name_vec with the file extension specified in this_ext. Default is no change in file entry in the spec file.
this_ext	Character string; file extension for data files. Default is ".dat".
this_title_list	List of character strings with the titles for each series. Default is to set title to the series name.
this_title_base	Character string; Optional base for custom title; series name will be added at the end of the title; quotes will be added by the routine. Default is to set title to the series name.
this_xreg_list	list of character strings with the filenames of user defined regressors or NULL for each series. Default is to not set regression.file for the individual series.

this_user_list
: List of vectors of character strings with the names of user defined regressors or NULL for each series. Default is to not set `regression.file` for the individual series.

make_metafile
: Logical scalar; if `TRUE`, generate a makefile for this set of files; do not otherwise. Default is `FALSE`.

this_metafile_name
: Character string; base name of metafile to be generated. If not specified, use name of list input as metafile name. Note: do not specify the ".mta" file extension.

this_meta_directory
: Character string; name of optional directory where the meta file is stored. If not specified, the metafile will be saved in the current working directory.

this_output_directory
: Character string; name of optional directory where the output files are stored in the metafile. If not specified, the output files are saved in the current working directory.

include_directory
: Logical scalar; if `TRUE`, include directory specified in `this_directory` with file name output. Otherwise, output only names in `this_name_vec`. Default is `FALSE`.

Details

Version 7.1, 2026-03-05

Value

Stores the spec file representation of each element of a list of `seas` objects into separate files.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list,
      slidingspans = "",
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      regression.aictest = NULL,
      forecast.maxlead = 36,
      check.print = c( "pacf", "pacfplot" ))
test_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
## Not run: save_spec_file_vec(test_seas_update,
             this_name_vec = c('n3000013', 'n3000014', 'n3000025', 'n3000026'),
             this_data_directory = 'X:\\seasonalAdj\\testing\\data',
             this_directory = 'X:\\seasonalAdj\\testing\\sautilities',
             this_meta_directory = 'X:\\seasonalAdj\\testing\\sautilities',
             this_title_base = 'Production Run for US Unemployment : ',
             make_metafile = TRUE, include_directory = TRUE)
## End(Not run)
```

`save_ts_to_csv` *save a ts object into a csv file ready to be read by Python*

Description

Save a `ts` object into a `.csv` file that is ready to be read by Python.

Usage

```
save_ts_to_csv(this_ts = NULL, this_csv_file = NULL, start_date = NULL)
```

Arguments

`this_ts`	Double precision time series array to be saved. This is a required argument.
`this_csv_file`	Character string; name of file time series array to be saved to. This is a required argument.
`start_date`	Integer vector of length 2, Start date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is start of series.

Details

Version 1.4, 2026-07-01

Value

File with time series will be produced.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
UKgas_tc_date <- c(1970, 2)
UKgas_tc_1970_2 <-
   gen_tc_outlier_ts(UKgas_tc_date,
      this_start = c(1960, 1),
      this_end = c(1990, 4),
      this_freq = 4,
      tc_alpha = 0.9,
      return_matrix = FALSE)
## Not run: save_ts_to_csv(UKgas_tc_1970_2, this_csv_file = 'UKgas_tc_1970_2.csv')
```

save_user_reg *save user regression matrix*

Description

Save a user-defined regression array or matrix with time series attributes to an external ASCII file in X-13ARIMA-SEATS' datevalue format. For more information on the datevalue format of X-13ARIMA-SEATS, see Monsell et al. (2016).

Usage

```
save_user_reg(
  this_reg = NULL,
  this_reg_file = NULL,
  add_dates = TRUE,
  start_date = NULL
)
```

Arguments

this_reg	Double precision time series array or matrix to be saved. This is a required argument.
this_reg_file	Character string; name of file time series array or matrix to be saved to. This is a required argument.
add_dates	Logical scalar; save the file with dates on each record (datevalue format). Default is TRUE.
start_date	Integer vector of length 2, Start date for series to be stored, where the first element is the year and the second element is the month or quarter. Default is start of series.

Details

Version 4.0, 2026-03-04

Value

File with user-defined regressors will be produced.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Monsell BC, Lytras D, Findley D (2016). "Getting Started with X-13ARIMA-SEATS Input Files." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/gettingstartedx13-winx13.pdf.

Examples

```
UKgas_tc_date <- c(1970, 2)
UKgas_tc_1970_2 <-
   gen_tc_outlier_ts(UKgas_tc_date,
      this_start = c(1960, 1),
      this_end = c(1990, 4),
      this_freq = 4,
      tc_alpha = 0.9,
      return_matrix = TRUE)
## Not run: save_user_reg(UKgas_tc_1970_2, 'UKgas_tc_1970_2.txt')
```

`seasonal_ftest` *Model-based F-Test for Time Series Models.*

Description

Model based test for seasonality based on stable seasonal regressors. For more information on the model based seasonal F-test with applications, see Bell et al. (2022) and Monsell (2024).

Usage

```
seasonal_ftest(
  seas_obj = NULL,
  p_limit_fail = 0.01,
  p_limit_warn = 0.05,
  use_seasonal = TRUE,
  return_this = "test"
)
```

Arguments

`seas_obj`	A `seas` object for a single series generated from the `seasonal` package (Sax and Eddelbuettel 2018). Note that either seasonal or sine-cosine trigononmetric regressors must be included in the regARIMA model. This is a required entry.
`p_limit_fail`	Numeric scalar; P-value limit for model based seasonal F-statistic for passing. Default is `0.01`.
`p_limit_warn`	Numeric scalar; P-value limit for model based seasonal F-statistic for a warning. Default is `0.05`.
`use_seasonal`	Logical scalar; if TRUE, the `seasonal` regressor is used; otherwise, use the sine-cosine trignonmetric regressors generated by `sincos`.
`return_this`	Character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 6.0, 2026-03-04

Value

A text string denoting if the series passed or failed tests for seasonality using the model based F-test diagnostic.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Monsell BC (2024). "Two Topics in Applying Seasonal Adjustment Diagnostics on Large Sets of Series." https://www.bls.gov/osmr/research-papers/2024/pdf/st240110.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
m_air <-
  seasonal::seas(AirPassengers,
     transform.function = "log",
     arima.model = '(0 1 1)',
     regression.variables = c('seasonal', 'td'),
     x11="")
this_seasonal_ftest <-
   seasonal_ftest(m_air, return_this = 'both')
```

---

set_critical_value *Set outlier critical value*

Description

Set outlier critical value using the Ljung algorithm as given in Ljung (1993).

Usage

```
set_critical_value(number_observations = NULL, cv_alpha = 0.01)
```

Arguments

number_observations	Integer scalar; number of observations tested for outliers. This is a required entry.
cv_alpha	Numeric scalar; alpha for critical value. Default is 0.01.

Details

Version 2.0, 2026-03-04

Value

Outlier critical value generated by the algorithm given in Ljung (1993). The critical value in X-13ARIMA-SEATS is different as it is adjusted to allow for smaller values to approximate the normal distribution.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM (1993). "On Outlier Detection in Time Series." *Journal of Royal Statistical Society B*, **55**, 559–567. doi:10.1111/j.25176161.1993.tb01924.x.

Examples

```
this_critical_value <- set_critical_value(12, 0.025)
```

`set_day_and_month` *generate date object from days and year*

Description

Generate a date object from the number of days from the beginning of the year and the target year. The `lubridate` package is used in this function (Grolemund and Wickham 2011).

Usage

```
set_day_and_month(day_number = NULL, target_year = NULL)
```

Arguments

`day_number`	Integer scalar, number of days from the start of the year. This is a required entry.
`target_year`	Integer scalar, year of the date object. This is a required entry.

Details

Version 2.0, 2026-03-04

Value

Date object determined by the input arguments.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). "Dates and Times Made Easy with lubridate." *Journal of Statistical Software*, **40**(3), 1–25. https://www.jstatsoft.org/v40/i03/.

Examples

```
this_start <- set_day_and_month(1, 2001)
this_end   <- set_day_and_month(350, 2025)
```

set_legend_position *generate position of plot legend*

Description

Generate position code for the legend command based on the series being plotted.

Usage

```
set_legend_position(
  data_matrix = NULL,
  this_plot_start = NULL,
  this_plot_freq = 12,
  time_disp = 3,
  value_disp = 1/6,
  default_code = "top"
)
```

Arguments

data_matrix	numeric matrix; Matrix where all series being plotted are stored as columns. This is a required entry.
this_plot_start	Integer scalar; start date of the plot. This is a required entry.
this_plot_freq	Integer scalar; Frequency of time series plotted. Default is 12.
time_disp	Integer scalar; number of observations on the x-axis taken up by the legend. Default is 3.
value_disp	Numeric scalar; factor representing the percentage of the y axis taken up by the legend. Default is 1/6.
default_code	Character string; default position code if the corners are not available. Possible values are "bottomright", "bottom", "bottomleft", "left", "topleft", "topright", "top", "right" and "center". Default is "top".

Details

Version 2.9, 2026-03-04

Value

Position codes for the legend command. Possible values are "bottomright", "bottom", "bottomleft", "topleft", "topright" and the value of default_code.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
shoes_seas <-
   seasonal::seas(shoes2007,
      transform.function = "log",
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead=36,
      x11 = "",
      slidingspans = "")
this_matrix <- cbind(shoes2007, seasonal::final(shoes_seas))
this_legend_position <-
   set_legend_position(this_matrix,
      this_plot_start = start(shoes2007),
      this_plot_freq = frequency(shoes2007),
      time_disp = 8,
      value_disp = 1/8,
      default_code = "top")
```

---

shoes2007 *Retail sales of shoes, ending in 2007*

Description

A time series object of retail sales of shoes published by the US Census Bureau ending in December of 2007.

Usage

```
shoes2007
```

Format

A `ts` object of retail sales of shoes ending in December of 2007.

---

shoes2008 *Retail sales of shoes, ending in 2008*

Description

A time series object of retail sales of shoes published by the US Census Bureau ending in April of 2008.

Usage

```
shoes2008
```

Format

A `ts` object of retail sales of shoes ending in April of 2008.

significant_t_stats *Number of signifcant uregressors*

Description

Determines the number of regressors in a seas object that are significant.

Usage

```
significant_t_stats(seas_obj = NULL, critical_value = 2, user_def_only = FALSE)
```

Arguments

seas_obj	Object generated by seas() of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
critical_value	Numeric scalar; sets the critical value used to test t-statistics of regressors. Default is 2.0.
user_def_only	Logical scalar; only test user-defined regressors Default is FALSE.

Details

Version 3.0, 2026-03-04

Value

Number of regressors with significant t-statistics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      regression.aictest = "td",
      forecast.maxlead=36,
      check.print = c( "pacf", "pacfplot" ),
      check.save = "acf",
      x11="",
      slidingspans = "")
air_num_user_reg <- significant_t_stats(air_seas)
```

spec_peak_fail_why *Failure text for spectral peaks*

Description

Generate text on why spectral peaks are flagged. For more information on spectral plots in X-13ARIMA-SEATS, see Section 6.1 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
spec_peak_fail_why(
  udg_list = NULL,
  peak_level = 6,
  this_spec = "spcsa",
  return_both = FALSE
)
```

Arguments

udg_list	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
peak_level	Integer scalar - limit to determine if a frequency has a spectral peak. Default is `6`.
this_spec	Text string with the spectrum being tested allowable entries are `'spcori'`, `'spcsa'`, `'spcirr'`, `'spcrsd'`. Default is `'spcori'`.
return_both	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 4.2, 2026-03-05

Value

A text string denoting if the series passed the tests of spectrum diagnostics, or why the series did not pass. Note that for `'spcori'`, the series fails if none of the frequencies tested had peaks. For more information on spectral peak detection in X-13ARIMA-SEATS, see Bell et al. (2022) and Soukup and Findley (1999).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of*

Statistical Software, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). "On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment." *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. `https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf`.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, `https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf`.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      x11 = "")
air_seas_udg <- seasonal::udg(air_seas)
this_spec_peak_fail_why <-
   spec_peak_fail_why(air_seas_udg,
      this_spec = 'spcori',
      return_both = TRUE)
```

`spec_peak_test` *Test for spectral peaks*

Description

Test if spectral peaks are flagged by X-13ARIMA-SEATS. For more information on spectral plots, see Section 6.1 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Usage

```
spec_peak_test(
  seas_obj = NULL,
  peak_level = 6,
  peak_warn = 3,
  this_spec = "spcsa",
  return_this = "test"
)
```

Arguments

`seas_obj`	Object generated by `seas()` of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`peak_level`	Integer scalar - limit to determine if a frequency has a spectral peak. Default is `6`.
`peak_warn`	Integer scalar - limit to produce a warning that a frequency may have a spectral peak. Default is `3`.
`this_spec`	text string with the spectrum being tested. Allowable entries are `'spcori'`, `'spcsa'`, `'spcirr'`, `'spcrsd'`. Default is `'spcori'`.

return_this   character string; what the function returns - 'test' returns test results, 'why' returns why the test failed or received a warning, or 'both'. Default is 'test'.

Details

Version 5.2, 2026-03-05

Value

A text string denoting if the series passed or failed the tests of spectrum diagnostics. Note that for spcori, the series fails if none of the frequencies tested had peaks. For more information on spectral peak detection in X-13ARIMA-SEATS, see Bell et al. (2022) and Soukup and Findley (1999).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). "On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment." *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      x11 = "")
this_spec_peak_test <-
   spec_peak_test(air_seas,
      this_spec = 'spcori',
      return_this = 'both')
```

spec_peak_warn_why *Warning message for spectral peaks*

Description

Generate warning message related to spectral peaks for a given X-13ARIMA-SEATS spectrum. For more information on spectral plots in X-13ARIMA-SEATS, see Section 6.1 of U. S. Census Bureau (2025).

Usage

```
spec_peak_warn_why(
  udg_list = NULL,
  peak_warn_level = 3,
  this_spec = "spcsa",
  return_both = FALSE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`peak_warn_level`	Integer scalar - limit to produce a warning that a frequency may have a spectral peak. Default is `3`.
`this_spec`	text string with the spectrum being tested. Allowable entries are `'spcori'`, `'spcsa'`, `'spcirr'`, `'spcrsd'`. Default is `'spcori'`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `FALSE`.

Details

Version 4.3, 2026-03-09

Value

A text string denoting if the series passed the tests of spectrum diagnostics, or why the series did not pass. Note that for `spcori`, the series fails if none of the frequencies tested had peaks. For more information on spectral peak detection in X-13ARIMA-SEATS, see Bell et al. (2022) and Soukup and Findley (1999).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Bell W, McDonald-Johnson K, McElroy T, Pang O, Monsell B, Chen B (2022). "Identifying Seasonality." BLS Research Reports, U.S. Bureau of Labor Statistics, https://www.bls.gov/osmr/research-papers/2022/pdf/st220010.pdf.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). "On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment." *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      x11 = "")
air_seas_udg <- seasonal::udg(air_seas)
this_spec_peak_warn_why <-
   spec_peak_warn_why(air_seas_udg,
      this_spec = 'spcori',
      return_both = TRUE)
```

`sspan_test` *Sliding Spans Diagnostic*

Description

Results from the sliding spans diagnostic. For more information on sliding spans analysis, see Section 6.2 and 7.16 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025) and Findley et al. (1990).

Usage

```
sspan_test(
  seas_obj = NULL,
  sf_limit = 25,
  change_limit = 40,
  additivesa = FALSE,
  return_this = "test"
)
```

Arguments

`seas_obj`	Object generated by `seas()` of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`sf_limit`	Numeric object; limit for the percentage of seasonal spans flagged. Default is `25`.
`change_limit`	Numeric object; limit for the percentage of month-to-month changes flagged. Default is `40`.

`additivesa` Logical scalar; if `TRUE`, the adjustment is assumed

`return_this` Character string; what the function returns - `'test'` returns test results, `'why'` returns why the test failed or received a warning, or `'both'`. Default is `'test'`.

Details

Version 4.2, 2026-03-05

Value

A text string denoting if the series passed or failed the tests generated by the sliding spans diagnostics.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead=20,
      x11="",
      slidingspans = "")
ukgas_sspan_test <-
   sspan_test(ukgas_seas,
      sf_limit = 15,
      change_limit = 35,
      return_this = 'both')
```

`sspan_test_why` *Sliding Spans Diagnostic Warning Messages*

Description

Generate text on why tests using the sliding spans diagnostic fail. For more information on sliding spans analysis, see Section 6.2 and 7.16 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025) and Findley et al. (1990).

Usage

```
sspan_test_why(
  udg_list = NULL,
  sf_limit = 25,
  change_limit = 40,
  additivesa = FALSE,
  return_both = FALSE
)
```

Arguments

`udg_list`	List object generated by `udg()` function of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`sf_limit`	Numeric object; limit for the percentage of seasonal spans flagged. Default is `25`.
`change_limit`	Numeric object; limit for the percentage of month-to-month changes flagged. Default is `40`.
`additivesa`	logical scalar; if true, the adjustment is assumed to be additive. Default is `FALSE`.
`return_both`	Logical scalar indicating whether the calling function will return both the test results and why the test failed or produced a warning. Default is `TRUE`.

Details

Version 5.2, 2026-03-05

Value

A text string denoting if the series passed the tests of sliding spans diagnostics, or why the series failed.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
ukgas_seas <-
   seasonal::seas(UKgas,
      series.period = 4,
      arima.model = "(0 1 1)(0 1 1)",
      transform.function = "log",
      check.print = c( "pacf", "pacfplot" ),
      forecast.maxlead=20,
      x11="",
      slidingspans = "")
ukgas_seas_udg <- seasonal::udg(ukgas_seas)
ukgas_sspan_test_why <-
   sspan_test_why(ukgas_seas_udg,
      sf_limit = 15,
      change_limit = 35,
      return_both = TRUE)
```

`static_with_outlier` *Add outlier options and updated data to* `seas` *object*

Description

Add arguments from the `outlier` spec to a `seas` object. This function uses the `static` function from the `seasonal` package; for more information on generating and evaluating a static seas object, see Section 8.1 of Sax and Eddelbuettel (2018).

Usage

```
static_with_outlier(
  seas_obj = NULL,
  new_data = NULL,
  outlier_span = ",",
  outlier_types = "ao,ls",
  outlier_method = "addone",
  outlier_critical = NULL,
  cv_alpha = 0.01
)
```

Arguments

`seas_obj`	Object generated by `seas()` of the `seasonal` package (Sax and Eddelbuettel 2018). This is a required entry.
`new_data`	Time series object; updated data set from the data used to generate `seas_obj`. This is a required entry.

outlier_span character string; sets the argument outlier.span. Default is ",".

outlier_types character string; sets the argument outlier.types. Default is "ao,ls".

outlier_method Character string; sets the argument outlier.method. Default is "addone".

outlier_critical
Numeric scalar; sets the argument outlier.critica. Default is Ljung choice for number of observations in new_data. (Ljung 1993).

cv_alpha Numeric scalar; alpha for critical value. Default is 0.01.

Details

Version 3.2, 2026-03-05

Value

An updated static seas object with outlier arguments included. For more information about arguments to the outlier spec, see Section 7.11 of the X-13ARIMA-SEATS Reference Manual (U. S. Census Bureau 2025).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM (1993). "On Outlier Detection in Time Series." *Journal of Royal Statistical Society B*, **55**, 559–567. doi:10.1111/j.25176161.1993.tb01924.x.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
shoes_seas <-
   seasonal::seas(shoes2007,
      transform.function = "log",
      forecast.maxlead = 60,
      x11 = "")
shoes_seas_outlier <-
   static_with_outlier(shoes_seas,
      new_data = shoes2008,
      outlier_types = 'all',
      outlier_critical = 4.0)
```

`static_with_outlier_list` *Add outliers options and updated data to a list of* `seas` *object*

Description

Add outlier arguments and updated data to each element of a list of `seas` objects . This function uses the `static` function of the `seasonal` pacakge; for more information on generating and evaluating a static seas object, see Section 8.1 of Sax and Eddelbuettel (2018). Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
static_with_outlier_list(
  seas_obj_list = NULL,
  new_data_list = NULL,
  outlier_span = ",",
  outlier_types = "ao,ls",
  outlier_method = "addone",
  outlier_critical = NULL,
  cv_alpha = 0.01
)
```

Arguments

`seas_obj_list`	List object of `seas` objects generated from a call of `seas` on a list consisting of individual time series. This is a required entry.
`new_data_list`	List object of time series (`ts`) objects; updated data sets from the data used to generate `seas_obj_list`. This is a required entry.
`outlier_span`	Character string; sets the argument `outlier.span`. Default is `","`.
`outlier_types`	Character string; sets the argument `outlier.types`. Default is `"ao,ls"`.
`outlier_method`	Character string; sets the argument `outlier.method`. Default is `"addone"`).
`outlier_critical`	Numeric scalar; sets the argument `outlier.critica`. Default is Ljung choice for number of observations in `new_data`. (Ljung 1993).
`cv_alpha`	Numeric scalar; alpha for critical value. Default is `0.01`.

Details

Version 5.0, 2026-03-09

Value

A list of updated static seas object with outlier arguments included. For more information about arguments to the `outlier` spec, see Section 7.11 of the X-13ARIMA-SEATS Reference Manual U. S. Census Bureau (2025).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Ljung GM (1993). "On Outlier Detection in Time Series." *Journal of Royal Statistical Society B*, **55**, 559–567. doi:10.1111/j.25176161.1993.tb01924.x.

Sax C (2024). "Adjusting Multiple Series — cran.r-project.org." [Accessed 20-02-2026], `https://cran.r-project.org/web/packages/seasonal/vignettes/multiple.html`.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, `https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf`.

Examples

```
xt_seas_old <-
   seasonal::seas(xt_data_old,
      transform.function = "log",
      forecast.maxlead = 60,
      x11 = "")
xt_outlier_seas <-
   static_with_outlier_list(xt_seas_old,
      new_data_list = xt_data_new,
      outlier_types = "all",
      outlier_critical = 4.0)
```

`udg_series` *Process a list of* `seas` *elements*

Description

Process a list of `seas` objects generated by the `seasonal` package to find the elements that are greater than or less than a particular limit for a diagnostic. Information on generating adjustments of multiple time series stored in a list object with the `seasonal` package can be found in Sax (2024).

Usage

```
udg_series(
  seas_obj_list = NULL,
  this_key = "autoout",
  this_limit = 5,
  this_abs = FALSE,
  this_position = 1,
  greater_than = TRUE
)
```

Arguments

`seas_obj_list` List object of seas objects generated from a call of seas on a list consisting of individual time series. This is a required entry.

this_key	Character string containing keyword of the udg function of the seasonal package that returns a numeric value. For more information on the udg function, see Sax (2018). Default is 'autoout'.
this_limit	Numeric object which serves as the limit of the diagnostic referred to in this_key. Default is '5'
this_abs	Logical scalar that indicates whether the absolute value is taken of the numbers before the comparison is made. Default is FALSE.
this_position	Integer scalar; if the result is a vector, the position of number that the comparison is made to. Default is 1.
greater_than	Logical scalar that specified whether the element names returned are greater than (TRUE) or less than (FALSE) the limit specified in this_limit. Default is TRUE.

Details

Version 4.0, 2026-03-04

Value

A vector of list element names where this_key is greater than or less than the limit specified in this_limit. If nothing matches, the function will output the string 'none'.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
emp_seas_list <-
   seasonal::seas(employment_data_mts,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead=36,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "",
      slidingspans = "")
emp_seas_update <-
   Filter(function(x) inherits(x, "seas"), emp_seas_list)
xt_bad_m7 <-
   udg_series(emp_seas_update, this_key = 'f3.m07', this_limit = 1.2)
xt_bad_q2 <-
   udg_series(emp_seas_update, this_key = 'f3.qm2', this_limit = 1.2)
```

---

unemployment_data_mts *US Unemployment Series, four main components in an* mts *object*

Description

An mts object of the four main components of US Unemployment from the US Bureau of Labor Statistics expressed as time series objects that end in December, 2022

Usage

```
unemployment_data_mts
```

Format

An `mts` object with 4 time series elements in four columns:

`n3000013` Unemployed Males 16-19

`n3000014` Unemployed Females 16-19

`n3000025` Unemployed Males 20+

`n3000026` Unemployed Females 20+

`unemployment_list` *US Unemployment Series, four main components in a list object*

Description

Object of the four main components of US Unemployment from the US Bureau of Labor Statistics expressed as time series objects that end in December, 2022

Usage

```
unemployment_list
```

Format

A list object with 4 time series elements:

`n3000013` Unemployed Males 16-19

`n3000014` Unemployed Females 16-19

`n3000025` Unemployed Males 20+

`n3000026` Unemployed Females 20+

`update_diag_matrix` *Update Diagnostic Matrix*

Description

Update the matrix of diagnostics used to generate the diagnostic data frame in `make_diag_df`.

Usage

```
update_diag_matrix(
  this_diag_list = NULL,
  this_test_list = NULL,
  this_label = NULL
)
```

Arguments

- `this_diag_list` List object with elements for seasonal adjustment or modeling diagnostic, titles, and the number of columns. This is a required entry.
- `this_test_list` List object of a specific seasonal adjustment or modeling diagnostic. This is a required entry.
- `this_label` Character string; name of diagnostic in `this_test_list`. This is a required entry.

Details

Version 4.4, 2026-03-04

Value

List object with updated elements for seasonal adjustment or modeling diagnostic, titles, and the number of columns.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      regression.aictest = NULL,
      forecast.maxlead=60,
      check.print = c( "pacf", "pacfplot" ),
      x11 = "",
      slidingspans = "")
test_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
test_acf <-
   lapply(test_seas_update, function(x) try(acf_test(x, return_this = 'both')))
test_names <- names(xt_data_new)
num_names <- length(test_names)
all_diag_list <- list(n = 0, diag = 0, titles = 0)
if (!is.null(test_acf)) {
    if (length(test_acf) < num_names) {
       this_acf_test <-
          fix_diag_list(test_acf, test_names, return_this = 'both')
    }
    all_diag_list <-
       update_diag_matrix(this_diag_list = all_diag_list,
          this_test_list = test_acf, this_label = "ACF")
}
```

update_fts *Update final t-statistics from outlier identification*

Description

Update final t-statistics from automatic outlier identification with t-statistics from identified outliers in final regARIMA model. For more information on automatic outlier identification in X-13ARIMA-SEATS, see Sectiona 4.6 and 7.11 of the X-13ARIMA-SEATS Reference Manual U. S. Census Bureau (2025).

Usage

```
update_fts(seas_obj = NULL)
```

Arguments

seas_obj	A seas object generated from a call of seas on a single time series (Sax and Eddelbuettel 2018). This is a required entry.

Details

Version 2.1, 2026-03-09

Value

Updated fts matrix.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf.

Examples

```
air_seas <-
   seasonal::seas(AirPassengers,
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 60,
      forecast.save = "fct",
      outlier.save = "fts")
air_fts <- seasonal::series(air_seas, "fts")
air_fts_updated <- update_fts(air_seas)
```

update_vector *Update vector.*

Description

Fill unspecified elements of a vector with the first element of the input series.

Usage

```
update_vector(this_series = NULL, this_num = NULL)
```

Arguments

this_series	Original time series. This is a required entry.
this_num	Length of updated series. Must be more than the length of `this_series`. This is a required entry.

Details

Version 2.4, 2026-03-04

Value

An updated vector of length `this_num` augmented with the first value of the input series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_vector <- c(1,2)
updated_vector <- update_vector(this_vector, 4)
```

which_error *Check list for try errors*

Description

Checks list for try errors, returning element names with `'try-error'`. For an example of using the `try` function when using the `seasonal` package on multiple series, see Section 8.2 in Sax and Eddelbuettel (2018).

Usage

```
which_error(this_list = NULL)
```

Arguments

this_list	List object which potentially contains `'try-error'` class objects.

Details

Version 3.1, 2026-03-05

Value

Vector of the names of list elements that are 'try-error' class objects. If the list contains no 'try-error' class objects, the function will return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
unemp_seas_list <-
   seasonal::seas(unemployment_list,
      transform.function = "log",
      arima.model = "(0 1 1)(0 1 1)",
      forecast.maxlead = 36,
      check.print = c( "pacf", "pacfplot" ),
      slidingspans = "")
unemp_seas_update <-
     Filter(function(x) inherits(x, "seas"), unemp_seas_list)
unemp_seas_errors <- which_error(unemp_seas_update)
```

---

xt_data_list   *US Building Permits*

Description

A list object with 12 components of US Building Permits published by the US Census Bureau expressed as time series objects.

Usage

```
xt_data_list
```

Format

A list object with 12 time series elements:

mw1u Midwest one family building permits

mwto Midwest total building permits

ne1u Northeast one family building permits

neto Northeast total building permits

so1u South one family building permits

`soto` South total building permits

`we1u` West one family building permits

`weto` West total building permits

`us1u` US one family building permits

`us24` US 2-4 family building permits

`us5p` US 5+ family building permits

`usto` US total family building permits

`xt_data_new` *US Building Permits, One Family Buildings (new)*

Description

A list object of US One family Building Permits for four regions published by the US Census Bureau expressed as time series objects that end in October, 2006.

Usage

```
xt_data_new
```

Format

A list object with 4 time series elements:

`mw1u` Midwest one family building permits

`ne1u` Northeast one family building permits

`so1u` South one family building permits

`we1u` West one family building permits

`xt_data_old` *US Building Permits, One Family Buildings (old)*

Description

A list object of US One family Building Permits for four regions published by the US Census Bureau expressed as time series objects that end in December, 2005.

Usage

```
xt_data_old
```

Format

A list object with 4 time series elements:

`mw1u` Midwest one family building permits

`ne1u` Northeast one family building permits

`so1u` South one family building permits

`we1u` West one family building permits

Index